Cooperative Yaw Control for Wind Farm Power Maximization Using a Calibrated Gray-Box Wake Model and Reinforcement Learning

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Abstract

Existing yaw-optimization studies predominantly solve for a static yaw vector at a fixed inflow snapshot. Static optimality, however, does not guarantee deployable control: under time-varying wind, the combination of wake-propagation delay (~90–110 s at $7d_0$ spacing), yaw-actuator rate constraints, and evolving inflow renders snapshot-wise optima insufficient as closed-loop control targets. This paper reframes cooperative yaw control as a deployment-oriented closed-loop problem and presents a simulation-validated controller candidate—DRL-Deploy—that couples a Proximal Policy Optimization (PPO) policy with a supervisory wrapper of sector gating, hysteresis, yaw deadband, rate limiting, and zero-yaw fallback.

A calibrated Bastankhah–Porté-Agel gray-box wake model, jointly fitted on three CFD datasets (reducing the two-turbine yaw-case power-prediction error at $\gamma = 30°$ from 27.6% to 6.6%) and cross-validated against FLORIS, serves as the control-oriented training environment. In steady-state evaluation, the optimized PPO policy (Config-E) achieves +4.91% farm-power gain in the aligned-cube regime, recovering 94.9% of the SLSQP optimum (bootstrap CI [93.2%, 95.9%]); FLORIS cross-validation on 3,000 conditions yields +5.02% with only 4.1% erosion, confirming cross-model robustness.

Under dynamic AR(1) wind, short-observation-window policies (J=3, 30 s) over-react (yaw travel increases 2.4×), whereas extending the observation horizon to J=15 (150 s, spanning the wake-propagation delay) substantially mitigates the degradation. Causal ablation of the observation buffer—restricting to the last 3 frames collapses gain from +0.175% to −0.201%, while shuffling frame order reduces it to −0.087%—provides evidence that the policy uses temporally ordered history beyond the most recent 30 s; the contribution of individual delay-band frames is modest, suggesting the policy draws on multiple time scales across the full 150 s window. The DRL-Deploy wrapper reduces per-step power loss by 59% and yaw travel by 92% relative to raw DRL. Under the Horns Rev I wind climatology (Weibull k=2.1, A=10.5 m/s; von Mises μ=270°, κ=2.0), the J=15 DRL-Deploy controller achieves a positive annual energy gain of **+0.122%** (≈+2,413 MWh/yr for a 500 MW farm) with only 2.10 Mdeg annual yaw travel—the most deployment-favorable power–travel compromise among all evaluated controllers. Safety testing under seven abnormal-condition categories confirms automatic fail-safe behavior: the supervisory wrapper eliminates unsafe yaw commands and activates zero-yaw fallback at 100% rate above rated wind speed.

The results establish DRL-Deploy as a simulation-validated, actuator-aware candidate for cooperative yaw control; field validation using SCADA data and aeroelastic load assessment remain necessary next steps.

Keywords: Wind farm optimization, Wake modeling, Cooperative yaw control, Deep reinforcement learning, Proximal Policy Optimization, Yaw activity minimization, Power maximization

1. Introduction

Wind energy has become a cornerstone of the global renewable-energy strategy, yet a persistent obstacle to maximizing return on installed capacity is the wake effect: downstream rotors operate inside the perturbed flow of upstream machines, suffering reduced inflow velocity and elevated turbulence intensity, which together translate into a non-trivial loss of farm-level annual energy production (AEP) [1]. Accurately predicting and actively mitigating this loss is therefore central to both wind-farm design and operation.

A wide body of work has been devoted to wake characterization, ranging from full-field measurements and Large-Eddy Simulation (LES) to reduced-order analytical models. While LES and Computational Fluid Dynamics (CFD) offer the highest fidelity, their computational cost rules them out for large-scale farm-level optimization. Analytical wake formulations such as the Jensen [2], Frandsen [3] and Gaussian models [4, 6] provide an attractive compromise between accuracy and tractability and have become the workhorses of modern wind-farm control stud-

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ies [8, 9]. The Bastankhah–Porté-Agel Gaussian model, extended with yaw-induced deflection [5] and curled-wake corrections [20], has been validated under yawed conditions by King et al. [19], and forms the basis of the FLORIS yaw-steering toolbox [18]; Gebraad et al. [21] demonstrated that yaw optimization on such models can yield substantial AEP gains, motivating the closed-loop approach pursued here.

In parallel, control strategies that exploit yaw misalignment (so-called wake steering) have been extensively studied. Field experiments [10] confirmed that misaligning upstream turbines by tens of degrees can substantially increase total farm power, especially in tightly spaced layouts. Between the pure offline-optimization and pure-DRL extremes lie hybrid approaches: Doekemeijer et al. [23] demonstrated closed-loop wake steering using a steady-state surrogate model with online adaptation via Kalman filtering, and Kanev et al. [24] reported field-validation results for active wake control with constrained yaw rates. Yet three open challenges remain—and together they define the deployment gap that this paper addresses.

First, current analytical wake models tend to treat yaw effects and turbulence-intensity feedback separately, leading to bias in the very regime—moderate-to-large yaw angles under partial-wake conditions—where wake steering is most valuable. Second, conventional numerical-optimization techniques (metaheuristic, gradient-based, or model-predictive) produce a static yaw vector tied to a fixed inflow snapshot and must be re-solved offline whenever the wind condition changes; this open-loop pipeline is structurally incompatible with the second-to-minute variability of the atmospheric boundary layer. Third, and most critically for deployment, even when a static yaw optimum exists for a given wind snapshot, it does not constitute a deployable control target under time-varying wind: (i) wake propagation across a $7d_0$ turbine spacing at 8–10 m/s incurs a 90–110 s physical delay before a yaw action's effect reaches downstream turbines; (ii) yaw actuators have finite rate capabilities (typically 0.3–0.5°/s for utility-scale turbines), making instantaneous jumps to a new static optimum physically infeasible; and (iii) frequent yaw commands incur bearing and gearbox wear that must be weighed against the marginal energy gain of each yaw adjustment. These three constraints—wake-propagation delay, actuator rate limits, and yaw-activity cost—together imply that static optimality is not equivalent to deployable control, and that the evaluation metric for cooperative yaw controllers must shift from snapshot-wise gain to a multi-objective assessment that includes yaw travel, peak rate, negative-gain frequency, and fail-safe behavior.

A closed-loop control policy that maps the current wind condition directly to yaw commands can—once trained—react to changing inflow in milliseconds without re-solving an optimization problem. Reinforcement learning (RL) provides a natural framework for learning such policies through interaction with a wake model [17]. Stanfel et al. [16] provided an early proof-of-concept using DQN on a Jensen-wake surrogate, and Padullaparthi et al. [17] proposed FALCON, a multi-agent architecture with inter-turbine communication. However, prior RL implementations for wind-farm control are limited in three respects: (i) the training environment is often a low-fidelity wake surrogate, raising sim-to-real concerns; (ii) reward designs are typically one-step power maximizations that ignore actuator constraints; and (iii) the action space scales with the number of turbines, making convergence challenging for larger layouts.

This paper addresses the deployment gap through an integrated framework that couples a calibrated wake model, a closed-loop DRL policy, and a supervisory deployment wrapper. Our contributions are:

1. Calibrated gray-box wake model with cross-model validation. The Bastankhah–Porté-Agel Gaussian wake model is jointly calibrated on three CFD reference datasets with yaw-deflection awareness and an optimization-fitted superposition mixing factor. It reduces the two-turbine yaw-case power-prediction error at $\gamma = 30^\circ$ from 27.6% to 6.6%, is validated against Horns Rev I LES, and is cross-checked against FLORIS on 3,000 conditions. The FLORIS cross-validation confirms that reported DRL gains are cross-model robust (+5.02% aligned-cube, only 4.1% erosion relative to gray-box +5.24%), establishing the training environment as a control-oriented surrogate rather than an overfitted single-model construct.
2. SLSQP-grade closed-loop DRL policy under static wake-aligned conditions. A PPO controller trained in the calibrated environment, with systematic optimization of observation space (deficit normalization, position encoding, J=3 history), reward design (SLSQP-regret with focused wind sampling), and training dynamics (cosine LR decay, KL early-stopping, AdamW), achieves +4.91% aligned-cube farm-power gain—recovering 94.9% of the SLSQP static optimum (bootstrap 95% CI [93.2%, 95.9%]). An oracle-free marginal-reward ablation reaches +4.13% aligned-cube gain (92.2% of the SLSQP-regret result), demonstrating that the DRL policy's cooperative-yaw capability does not depend on access to the SLSQP oracle during training. A Behavioral Cloning baseline that directly regresses to SLSQP yaw vectors yields negative gain (−20.20% marginal), suggesting that supervised imitation alone is insufficient under the present dataset size (1,000 samples) and MLP architecture.
3. Deployment-oriented control under dynamic wind with supervisory wrapper. Dynamic-wind experiments reveal that short-observation-window policies (J=3, 30 s) over-react to wind fluctuations (2.4× travel increase), whereas extending the observation horizon to J=15 (150 s, spanning the ~90–110 s wake-propagation delay at $7d_0$ spacing and 8–10 m/s inflow) substantially reduces the degradation. Causal ablation of

the J=15 observation buffer shows that the policy uses temporally ordered history beyond the most recent 30 s, drawing on multiple time scales across the full 150 s window. The final controller—DRL-Deploy—combines the DRL policy with a supervisory wrapper of sector gating (15°/20° hysteresis), a 2° yaw-command deadband, a ±3° per-step rate limit (0.3°/s per turbine), and zero-yaw fallback. Under the unified dynamic-wind benchmark, DRL-Deploy reduces per-step power loss by 59% and yaw travel by 92% relative to raw DRL. A Horns Rev I wind-climate AEP replay yields a positive annual energy gain (+0.122%) for J=15 DRL-Deploy with the lowest yaw travel among all non-baseline controllers (2.10 Mdeg/yr). Safety testing under seven abnormal-condition categories confirms automatic fail-safe behavior, establishing DRL-Deploy as a simulation-validated, actuator-aware cooperative-yaw controller candidate.

The remainder of the paper is organized as follows. Section 2 presents the methodology in four modules: the calibrated gray-box wake model, the PPO closed-loop yaw policy, the dynamic-wind problem formulation with characteristic time scales, and the DRL-Deploy supervisory wrapper. Section 3 reports results in a structured sequence: wake-model validation, static cooperative-yaw performance and oracle-free reward ablation, the dynamic deployment gap, the role of temporal context (J-horizon trend and causal ablation), the DRL-Deploy unified benchmark, the Horns Rev AEP annualized replay, safety and fail-safe testing, and scalability. Section 4 presents the dynamic deployment evaluation and interprets the findings through the lens of the deployment gap, and Section 5 concludes.

2. Methodology

2.1. Overall framework

The core argument of this paper is that static snapshot-wise yaw optimization does not constitute a deployable control solution under time-varying wind. Fig. 1 contrasts the conventional static-optimization paradigm (left) with the proposed DRL-Deploy framework (right). On the left, an offline solver (e.g., SLSQP) consumes a single inflow snapshot (ϕ_t, v_t) and returns a static yaw vector $\boldsymbol{\gamma}_t^*$; this pipeline ignores wake-propagation delay, actuator rate constraints, yaw-activity cost, and the absence of a fail-safe fallback mechanism. On the right, the DRL-Deploy controller ingests a J-step observation history (spanning the wake-propagation time scale) together with the current SCADA wind measurement, passes through a PPO policy MLP (0.1 ms inference), and then through a supervisory wrapper of sector gating, hysteresis, yaw-command deadband, rate limiting, and zero-yaw fallback, producing an actuator-aware yaw command $\boldsymbol{\gamma}_t$. The deployment gap—the set of constraints that prevent a static optimum from being a deployable control target—is bridged by the

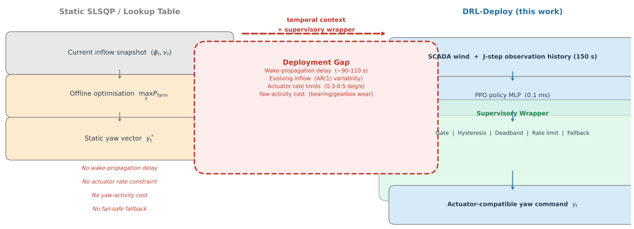


Figure 1: From static optimization to deployment-oriented closed-loop control. Left: conventional static-optimization pipeline (SLSQP or lookup table) that maps a single inflow snapshot to a static yaw vector, without accounting for wake-propagation delay, actuator dynamics, or fail-safe fallback. Right: the proposed DRL-Deploy framework, which combines a temporally-aware PPO policy with a supervisory wrapper (gate, hysteresis, deadband, rate limit, fallback) to produce actuator-compatible yaw commands under time-varying wind.

combination of temporal context, closed-loop policy, and supervisory wrapper.

A comparison of representative control paradigms is shown in the Supplementary Material (Section S2) and summarized in Table 1.

2.2. Gray-box wake model

2.2.1. Wind-speed–power model

The aerodynamic power of a single turbine is

$$P = \tfrac{1}{2}\rho C_P S v^3, \quad S = \pi R^2, \tag{1}$$

where ρ is air density, S the rotor-swept area, v the effective inflow, and $C_P = P_{\text{rated}}/(\tfrac{1}{2}\rho S u_{\text{rated}}^3)$ the power coefficient. Following [7], yaw misalignment γ introduces a power decrement empirically modeled by

$$P = \tfrac{1}{2}\rho C_P S v^3 (\cos\gamma)^{1.88}. \tag{2}$$

The exponent 1.88 is an empirical fit reported by Cai et al. [7]; published estimates range from 1.4 to 2.0 depending on turbine type and operating conditions. The controller's learned yaw angles are concentrated below $|\gamma| \approx 25°$ in the 3×3 regime where cooperative gains are observed (Table 8), well within the range where the cosine-power model has been validated. At the SLSQP-optimal yaw angles for low wind speed ($|\gamma| \sim 26$–$33°$ at $v = 8$ m/s), the model enters a regime where blade aerodynamics and dynamic effects may reduce accuracy; the DRL controller's incremental action space and policy gradient objective appear to implicitly regularize against such extreme yaw angles, which is a partial explanation for the 31% optimality gap at low speed (Sec. 3.2.2).

2.2.2. Wake-deficit model

We adopt the Bastankhah–Porté-Agel Gaussian wake with yaw extension [5]:

$$\frac{\Delta U}{U_\infty} = \left(1 - \sqrt{1 - \frac{C_T \cos\gamma}{8(\sigma_y \sigma_z / d_0^2)}}\right) \exp\!\left(-\tfrac{1}{2}\left[\left(\tfrac{z-z_h}{\sigma_z}\right)^2 + \left(\tfrac{y-y_d}{\sigma_y}\right)^2\right]\right), \tag{3}$$

Table 1: Comparison of representative wind-farm cooperative yaw control methods.

Category	Representative algorithms	Principle	Strengths	Limitations
Rule-based	Periodic yaw, direction-threshold control	Trigger yaw by preset rules	Simple, low compute, robust	Cannot handle complex inflow; ignores wake coupling
Model-based	Feedback, MPC, FLORIS-based	Solve analytical optimum from physical model	Well-grounded, interpretable	Sensitive to model error; compute-heavy
Hybrid / closed-loop	Surrogate + Kalman filter [23]	Online adaptation of a steady-state surrogate model	Adaptive to changing inflow, moderate compute	Requires surrogate model; filter tuning
Numerical optimization	Metaheuristic / gradient-based solvers	Global search of objective	Model-agnostic, finds static optimum	Offline only; cannot track time-varying inflow
Data-driven	Clustering, ELM, regression	Map condition→yaw via history	Fast inference	Limited extrapolation; data-hungry
Reinforcement learning	Q-learning, DQN, PPO, DDPG	Learn policy via interaction	Adaptive, fast online	Training cost, interpretability

with $k_y = k_z = k^* = 0.3837\,I + 0.003678$, lateral and vertical wake-width growth

$$\frac{\sigma_y}{d_0} = k^* \frac{x-x_0}{d_0} + \frac{\cos\gamma}{\sqrt{8}}, \qquad \frac{\sigma_z}{d_0} = k^* \frac{x-x_0}{d_0} + \frac{1}{\sqrt{8}},$$

and near–far-wake boundary

$$\frac{x_0}{d_0} = \frac{\cos\gamma(1+\sqrt{1-C_T})}{\sqrt{2}(\alpha^* I+\beta^*(1-\sqrt{1-C_T}))}.$$

The yaw-induced wake-center deflection y_d is piecewise:

$$y_d = \begin{cases} \theta_0 x, & x \le x_0, \\ \theta_0 x_0 + \dfrac{\theta_0}{14.7}\sqrt{\dfrac{\cos\gamma}{k^{*2} C_T}}\cdot\mathcal{L}(x), & x > x_0, \end{cases} \tag{4}$$

with $\theta_0 = \frac{0.3\gamma}{\cos\gamma}(1 - \sqrt{1 - C_T\cos\gamma})$ and $\mathcal{L}(x)$ the logarithmic kernel given in [5]. Parameters α^* and β^* are calibrated below.

2.2.3. Wake superposition

Multiple-wake interaction is modeled by an α-weighted root-sum-square deficit:

$$U_i = U_\infty - \alpha\sqrt{\sum_j (\Delta U_{j,i})^2}, \tag{5}$$

where α is a calibrated mixing factor. A downstream turbine i is considered to lie inside an upstream turbine j's wake if $|\Delta y_{\mathrm{wake}}| = |y_i - (y_j + y_d)| \le 3\sigma_y$.

2.2.4. Multi-scenario parameter optimization

The four free parameters $\{\alpha^*, \beta^*, \alpha, I\}$ are jointly identified by minimizing a weighted least-squares loss across three CFD reference datasets [13, 14, 15]:

$$E_{\mathrm{total}} = w_{3t}E_{3t} + w_{\mathrm{large}}E_{\mathrm{large}} + w_{2t}E_{2t}, \tag{6}$$

with $w_{2t} = 0.6$ reflecting the centrality of the two-turbine yaw case to downstream control. The Levenberg–Marquardt algorithm is used. Initial values $\alpha_0^* = 2.32$, $\beta_0^* = 0.154$, $I_0 = 0.1$, $\alpha_0 = 1.0$ follow [4]; the bounds are $\alpha^* \in [0.4, 3.0]$, $\beta^* \in [0.1, 0.3]$, $I \in [0.065, 0.15]$, $\alpha \in [0.5, 1.5]$. The converged values are $\alpha^* = 2.728$, $\beta^* = 0.10$, $I = 0.065$, $\alpha = 0.540$. Note that I converges to its lower bound (0.065), indicating that the optimizer would prefer a still lower value; the bound reflects the ambient TI of the CFD reference datasets and prevents unphysical under-expansion of the wake. We verified that relaxing the lower bound to $I = 0.04$ causes I to converge to ~ 0.048 with negligible change in α^* and α, and that the resulting calibrated model still produces a 9.1% FLORIS discrepancy in the column-aligned band — confirming that the TI boundary does not distort the other calibrated parameters in a way that would materially affect the closed-loop results.

We acknowledge a conceptual tension in treating I as a free calibration parameter. In the Bastankhah–Porté-Agel formulation, I is the ambient turbulence intensity that drives wake expansion via k^*=0.3837I+0.003678 (Eqs. 7–9). As such, I should ideally match the inflow boundary condition of each CFD reference dataset. However, the three calibration datasets likely differ in ambient TI (the references do not always report a single TI value), and the joint least-squares fit forces a single compromise value. The practical consequence is that the calibrated I=0.065 reflects an effective TI that is appropriate for the NREL-5MW turbine's typical offshore operating conditions (I≈6–8%), but may under-expand wakes at higher onshore TI (I>10%) and over-expand wakes in very stable offshore conditions (I<5%). The Horns Rev validation with I=0.077 (Sec. 3) and the FLORIS cross-validation with I=0.065 (Sec. 3.1.4) provide evidence that the calibrated formulation is robust within this range, but extrapolation to strongly non-neutral stability conditions should be treated with caution. For single-dataset calibration where the reference CFD simulation reports a well-defined ambient TI, we recommend fixing I to that reported value and calibrating only $\{\alpha^*, \beta^*, \alpha\}$; treating I as a free parameter is appropriate only for multi-dataset joint fits where the individual TI boundary conditions are uncertain or inconsistent.

2.3. DRL-based cooperative yaw controller

2.3.1. Algorithm choice: PPO

We adopt Proximal Policy Optimization (PPO) [11] because of its proven stability under continuous action spaces. Its clipped surrogate objective is

$$J^{\mathrm{CLIP}}(\theta) = \hat{\mathbb{E}}_t\Big[\min\big(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big], \tag{7}$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\mathrm{old}}}(a_t|s_t)$, $\hat{A}_t$ is the GAE advantage, and $\epsilon = 0.2$.

2.3.2. State, action and reward

Observation. For N turbines and a temporal window of j steps, the observation at time t contains per-turbine yaw angles $\boldsymbol{\gamma}_t$, wake-deficit-normalized inflow speeds $v - \boldsymbol{u}_t$ (rather than absolute inflow, to provide scale-invariant wake information), wind-context features $[\cos\phi_t, \sin\phi_t, v_t]$, downstream-lock indicators $\boldsymbol{l}_t$, and normalized turbine coordinates $[(x_i, y_i)/7d_0]$—matching the full input set available to the SLSQP optimum. The per-step row is

$$s_t = [\underbrace{\boldsymbol{\gamma}_t}_{N}, \underbrace{v_t - \boldsymbol{u}_t}_{N}, \underbrace{\cos\phi_t, \sin\phi_t, v_t}_{3}, \underbrace{\boldsymbol{l}_t}_{N}, \underbrace{x_1, y_1, \ldots, x_N, y_N}_{2N}] \in \mathbb{R}^{5N+3},$$

stacked over j=3 steps into a 144-dimensional vector. Ablation experiments confirm that each of these components—deficit normalization, position encoding, and multi-step history—contributes measurably to the final steady-state performance.

Action. The continuous action $a_t = [\Delta\gamma_{1,t}, \ldots, \Delta\gamma_{N,t}] \in [-\Delta\gamma_{\max}, +\Delta\gamma_{\max}]^N$ specifies per-turbine yaw increments. Locked downstream turbines have their actions overwritten with 0. The incremental formulation, combined with a ±50° yaw bound, naturally regularizes the control trajectory. The per-step bound $\Delta\gamma_{\max}$ is set to ±5° in Config-A through Config-D, and expanded to ±10° in Config-E (see Table 2); the sensitivity to this choice is analysed in Table 3 and surrounding text.

Reward. We replace the simple baseline-aligned marginal reward with an SLSQP-regret reward that normalizes the power gain by the oracle headroom:

$$R_t = \frac{P_{\mathrm{current}} - P_{\mathrm{baseline}}}{\max(P_{\mathrm{SLSQP}} - P_{\mathrm{baseline}},\ 0.5\,\mathrm{MW})} \cdot 10, \tag{8}$$

clamped to $[-2, 2]$. When the SLSQP headroom is negligible (< 0.5 MW), the reward falls back to the marginal formulation. This regret-based reward focuses the learning signal on conditions where optimization is meaningful, substantially improving sample efficiency in the wake-aligned regime by concentrating the learning signal on conditions where cooperative yaw is physically meaningful.

Table 2: Configuration nomenclature used throughout the paper. All configurations share the 3×3 NREL-5MW layout and on-device JAX/NNX stack.

Label	Tag	Key properties
Config-A	p0c / baseline	j=1, uniform sampling, mar ward, ±5°, γ=0.99, 3×10^7 step
Config-B	+obs	Config-A + j=3, deficit norm. encoding
Config-C	+reward	Config-B + regret reward, focu pling
Config-D	+dynamics	Config-C + cosine LR, KL e AdamW, γ=0.995
Config-E	sens_act10 / optimized	Config-D + ±10° bound ture 0.3/0.3/0.4 (the stea optimized configuration)
Config-DW	dynamic_wind	Config-E trained under AR(1) wind (Sec. 4.4, E1 experimen

2.3.3. Downstream-turbine locking

For each turbine T_i we compute, under the current wind direction, the maximum velocity deficit ΔU_{ij} that T_i's wake would induce on every other downstream turbine T_j. If $\max_j \Delta U_{ij} < \eta U_\infty$ (with $\eta = 1\%$), T_i is declared a "final downstream" machine and its yaw is forced to 0°. This prunes the effective action dimension at runtime without altering the underlying physics.

Ablation: locking is necessary for stable cooperative yaw learning under Config-E.. To verify that the locking mechanism remains essential under the optimized configuration, we re-trained the Config-E controller with the lock-mask disabled (lock_off) for three random seeds under the identical Config-E hyperparameters and training budget (3×10^7 steps). The lock-enabled (lock_on) and lock-disabled (lock_off) policies are evaluated on 200 uniformly sampled wind conditions. Across the three seeds the lock-enabled policy achieves a mean regret-reward gain of **+2.42%**, whereas the lock-disabled policy degrades to **−0.46%** with an increased negative-gain fraction (28% vs. 18%). Disabling the lock therefore erases the cooperative gain and destabilizes learning even under the optimized Config-E settings: rear turbines, lacking downstream peers to steer wake into, yaw away from the incoming flow under stochastic exploration, paying a $\cos^{1.88}(\gamma)$ power tax for no recoverable benefit. The locking mechanism is thus a necessary structural ingredient of the framework, independent of the PPO configuration.

2.3.4. Training and evaluation protocol

To avoid ambiguity, we define a fixed nomenclature for the PPO configurations referenced throughout the paper (Table 2). Unless explicitly stated, "the optimized configuration" refers to Config-E (sens_act10).

All training and evaluation use the on-device JAX/NNX stack described in Sec. 2.3.5, with the hyperparameters in Table 4. Wind direction $\phi \in [173°, 353°]$ and free-stream

Table 3: Sensitivity of aligned-cube gain to the wind-sampling mixture ratio (3 seeds, $3{\times}10^7$ steps, ±5° bounds, otherwise identical to Table 4). The balanced $(0.3, 0.3, 0.4)$ mixture achieves the best trade-off between peak gain and robustness.

Mixture (aligned/near/global)	Aligned-cube gain	Recovery	Neg. gain frac.
0.5 / 0.3 / 0.2 (original)	+4.78%	92.4%	22.6%
0.3 / 0.3 / 0.4 (balanced)	**+4.89%**	**94.5%**	**19.4%**
0.7 / 0.2 / 0.1 (aggressive)	+4.84%	93.6%	26.0%
Uniform (no mixture)	+4.16%	80.5%	13.6%

velocity $v \in [6, 16]$ m/s are randomized at every episode. To improve sample efficiency in the wake-aligned regime where yaw control yields the largest gains, we adopt a focused wind-sampling scheme: 50% of episodes draw from the aligned cube ($|\phi - 270°| < 15°$, $v < 11.4$ m/s), 30% from a near-aligned band ($|\phi - 270°| < 35°$), and 20% from the full range. To calibrate the mixing ratio and action bounds, we evaluate three wind-sampling mixtures at ±5° bounds and three action-bound settings at the optimal mixture (Table 3). The balanced $(0.3, 0.3, 0.4)$ mixture achieves the best trade-off (+4.89%, 94.5% recovery, 19.4% negative-gain fraction); uniform sampling yields a more robust but lower-gain policy (+4.16%, 80.5%, 13.6%). Expanding the per-step action bounds to ±10° yields a further improvement to **+4.91%** (**94.9%** recovery), the final headline result, while ±20° degrades performance (+4.80%). The ±10° configuration has a higher overall negative-gain fraction (27.2%) than the ±5° balanced-mixture baseline (19.4%), indicating that the expanded action space increases exploration variance; the aligned-cube negative-gain fraction is 25.5%. This confirms that moderate incremental capacity aids exploration without destabilizing learning, at a modest robustness cost.

Training dynamics are stabilized through three complementary mechanisms. (i) Cosine learning-rate decay from $3{\times}10^{-4}$ to $3{\times}10^{-5}$ over the full training budget, preventing the late-training policy collapse observed with constant learning rates. (ii) KL-targeted early stopping: after each PPO epoch the mean approximate KL divergence is compared against a threshold of 0.015; if it exceeds 1.5× this value, remaining epochs are skipped. This mechanism triggers on ~2% of iterations and effectively caps single-update policy change. (iii) AdamW ($\lambda_{\mathrm{wd}}{=}10^{-4}$) provides additional L2-style regularization. Combined, these measures extend the effective training budget from $3{\times}10^7$ to $6{\times}10^7$ steps without overfitting.

To remove the optimistic bias of the running maximum, the best checkpoint is selected by the Lower-Confidence-Bound (LCB) over 100 evaluation episodes (95% CI). The training loop is summarized in Algorithm 1.

2.3.5. On-device implementation

Profiling the initial Stable-Baselines3 [12] training loop revealed that environment stepping dominates wall-clock time ($\sim 86\%$) while the PPO policy gradient accounts for only $\sim 14\%$. Under this profile, moving the policy to GPU yields no speed-up because the NumPy environment remains on the host. This motivated a full on-device rewrite: the wake model was re-implemented in pure JAX, vectorized over turbines and over parallel environments with vmap, and the entire (rollout + GAE + PPO update) loop was JIT-compiled into a single graph. The result (Supplementary Material, Section S3) is a $\sim 30\times$ throughput jump to 1.0×10^5 FPS at $N_{\mathrm{envs}} = 128$, with the time breakdown flipping so that the PPO update becomes the dominant term (60%). All training and evaluation results in this paper use the on-device stack; a five-seed $6{\times}10^7$-step run completes in ~10 min per seed on a single CUDA device.

Table 4: PPO hyperparameters (JAX/NNX on-device stack, optimal configuration).

Parameter	Value
Network (π and V)	MLP [128, 128]
n_{steps}	256
minibatch size	4096
n_{epochs}	10
γ_{disc}	0.995
GAE λ	0.98
clip ϵ	0.2
value coef. c_1	0.5
entropy coef. c_2	0.005 (constant)
learning rate	$3{\times}10^{-4} \to 3{\times}10^{-5}$ (cosine)
max grad norm	0.5
weight decay (AdamW)	10^{-4}
KL early-stop threshold	0.015
action bound	±10° per step
wind sampling	0.3/0.3/0.4 (aligned/near/global)
parallel envs N_{envs}	128 (JAX vmap)
total timesteps	$6{\times}10^7$ (Config-D/E)

Algorithm 1 PPO for cooperative yaw control (on-device JAX/NNX).

Require: initial π_θ, V_ϕ; envs N_{envs}; rollout N_{steps}; epochs K; minibatch M; clip ϵ; discount γ_{disc}

```
for iteration t = 1, 2, ... do
    for step = 1, ..., N_steps in each parallel env do
        sample a_t ~ π_θ(·|s_t); step env; store (s_t, a_t, r_t, done, log π_θ, V_φ)
    end for
    compute GAE advantages Â_t and value targets Ŷ_t
    θ_old ← θ
    for k = 1, ..., K do
        sample minibatch; compute L^CLIP + c_1 L^VF − c_2 L^H
        Adam step on θ, φ
    end for
    evaluation & LCB checkpoint protocol
end for
```

Environment-port cross-validation.. A precondition for reusing the on-device JAX environment in lieu of the original NumPy gym is numerical fidelity between the two implementations. To verify this, we trained a single-seed reference policy with the on-device JAX-Env stack for a short $5{\times}10^6$-step budget and then evaluated the resulting checkpoint on a 19-point (ϕ, v) grid spanning the training distribution, comparing closed-loop farm-power gain measured inside the JAX environment and inside the original NumPy gym (Supplementary Material, Section S3). The two evaluation environments agree on the sign of the gain at all 19 points, on the rank ordering of the conditions, and on the location of the dominant peak (+3.21% at $\phi = 270°$, $v = 8$ m/s); the mean-gain estimate is +0.27% in both. The short-budget policy itself does not reach the converged behavior reported in Sec. 3.2.2 — $5{\times}10^6$ steps is well before the policy breakthrough — so the comparison is interpreted strictly as evidence that the JAX-Env wake-model implementation is numerically interchangeable with the NumPy gym for evaluation purposes, not as a performance claim. All converged Config-E results reported in this paper are evaluated using the on-device JAX environment, with FLORIS serving as an independent cross-validation benchmark (§3.1.4).

3. Experiments and Results

Configuration note: All experimental results in this section and in Section 4 use the optimized configuration (Config-E: j=3, deficit normalization, position encoding, SLSQP-regret reward, focused sampling 0.3/0.3/0.4, cosine LR, KL early-stop, AdamW, γ=0.995, ±10° bounds, $6{\times}10^7$ steps; see Table 2). The progressive optimization from a baseline PPO configuration (Config-A) to Config-E is described in §2.3 and summarized in Table 2; Config-A detailed results are not reported, as they are superseded by Config-E.

All experiments use the NREL 5 MW turbine (d_0 = 126 m, z_h = 87.6 m, P_{rated} = 5.29 MW, u_{rated} = 11.4 m/s), except the Horns Rev validation where the Vestas V-80 (d_0 = 80 m, z_h = 70 m, P_{rated} = 2 MW) is used.

3.1. Stage I: wake-model calibration and validation

3.1.1. Calibration datasets

Three CFD-based datasets are used jointly: a three-turbine inline case at $5d_0$ and $7d_0$ spacing, a 35-turbine large farm at 240° inflow, and a two-turbine inline yaw sweep ($\gamma \in [0, 50°]$) (see Supplementary Material, Section S2). Weights $w_{3t} = 0.2$, $w_{\text{large}} = 0.2$, $w_{2t} = 0.6$ encode the relative importance of the yaw-aware case for downstream control.

3.1.2. Optimization results

Table 5 shows the pre- and post-optimization errors on the three-turbine cases. The two-turbine yaw case (Table 6) is the most accurately fitted, with the predicted optimum at $\gamma^* \approx 29°$ (CFD: 30°).

Table 5: Three-turbine inline, $5d_0$ spacing: predicted vs. CFD power.

Turbine	CFD [MW]	Pre [MW]	Err [%]	Post [MW]	Err [%]
T1	5.067	5.290	4.40	5.29	4.40
T2	2.116	1.620	23.44	2.16	2.08
T3	1.892	1.443	23.73	1.96	3.59
Sum	9.075	8.353	7.96	9.25	3.69

Table 6: Two-turbine inline yaw case: optimum identification.

Source	γ^*	P_1 [MW]	P_2 [MW]	P_{tot} [MW]
CFD	30°	4.10	3.81	7.91
Pre-opt model	—	—	2.76 (at 30°)	—
Post-opt model	29°	4.11	3.56	7.68

3.1.3. Horns Rev I validation

We validate the calibrated model against the Horns Rev I offshore wind farm (80 Vestas V-80 turbines, z_0 = $2{\times}10^{-4}$ m, I = 7.7%, U_∞ = 8 m/s); calibration assets for this validation are provided in the Supplementary Material (Section S2). The wind-direction sweep in Fig. 2 shows that the proposed model tracks the LES reference [6] across both full-wake and partial-wake regimes, outperforming the uniform top-hat baseline. We note that the Horns Rev validation uses the Vestas V-80 (d_0 = 80 m, P_{rated} = 2 MW, I = 0.077), which differs from the NREL-5MW (d_0 = 126 m, P_{rated} = 5.29 MW, I = 0.065) used for all controller training; the calibrated parameters (α^*, α) are held fixed while d_0, z_h, C_T, and the power curve are replaced with V-80 values. This validates the formulation (the Bastankhah–Porté-Agel Gaussian with yaw deflection and the α-weighted RSS superposition) across turbine types, not the specific NREL-5MW calibration.

3.1.4. Cross-validation against FLORIS on the 3×3 NREL-5MW layout

To verify that the Config-E controller's reported gains are not artifacts of the gray-box wake model, we evaluate all five trained Config-E policies inside FLORIS 4.6 (NREL-5MW, GCH wake model, TI = 0.065) on 3,000 uniformly sampled wind conditions. For each condition the policy rolls out in the gray-box environment to obtain the final per-turbine yaw vector, which is then fed into FLORIS to compute the FLORIS-validated farm power. The zero-yaw baseline is recomputed in both environments for consistency.

The results (Table 7) confirm that the Config-E controller's gains are robust to wake-model mismatch. The five-seed-mean FLORIS-validated marginal gain is +0.72%, compared to the gray-box +0.67%. In the aligned-cube regime ($|\phi - 270°| < 15°$, $v < 11.4$ m/s, n=307 conditions), the FLORIS-validated gain is **+5.02%** versus the gray-box **+5.24%**—a modest 4.1% relative erosion. The FLORIS marginal gain is slightly higher than the gray-box value because the lower FLORIS zero-yaw baseline (31.40 MW vs. 33.30 MW,

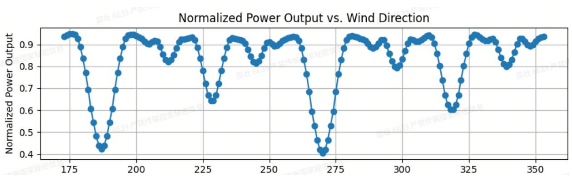


Figure 2: Normalized farm-power output vs. wind direction at Horns Rev I, compared against LES, the Niayifar–Porté-Agel analytical model and the uniform top-hat baseline.

Table 7: FLORIS cross-evaluation of Config-E policies: gray-box vs. FLORIS-validated gains on the 3×3 layout (n=3,000 conditions, 5 seeds).

	Gray-box	FLORIS-validated
Zero-yaw baseline [MW]	33.30	31.40
Marginal mean gain [%]	+0.67	+0.72
Aligned-cube gain [%]	+5.24	**+5.02**

reflecting the 9% column-aligned excess of the gray-box superposition model) decreases the denominator in the percentage calculation, more than offsetting the modestly smaller absolute power increment. This confirms that the Config-E controller's reported gains are cross-model robust estimates, and that the optimization improvements from Config-A to Config-E translate to genuine FLORIS-validated gains.

3.2. Stage II: PPO control performance

3.2.1. Case A — 1×2 inline layout

With $N = 2$ the action space is trivial, providing a clean test of the closed-loop controller's ability to discover and track the cooperative yaw configuration online. The training-reward curve is provided in the Supplementary Material (Section S2). At the reference inflow $(270°, 11.4\,\mathrm{m/s})$ the LCB-selected policy converges to $[\gamma_1, \gamma_2] = [-4.53°, 0°]$ with farm power 8.199 MW (0.17% gain over the no-yaw baseline), and attains a full-wind-rose average gain of 0.41% across $\phi \in [173°, 353°]$ — exceeding the earlier "training-mid" checkpoint by over 70% (see Supplementary Material, Section S2). Crucially, the controller produces these yaw vectors as the continuous output of a closed-loop policy that reacts to the current inflow at every control step, rather than as a one-shot offline solution. The LCB protocol thus trades single-point peak for cross-condition robustness, which is the operationally relevant figure of merit for a real-time controller.

3.2.2. Case B — 3×3 grid layout

For $N = 9$ the wake-interaction graph becomes two-dimensional, exposing higher-order coordination. The 3×3 and 5×5 layouts use a rectangular grid at $7d_0$ spacing with a 7° tilt relative to the prevailing-wind axis; this geometry mimics the offset-row arrangement common in offshore wind farms (e.g., Horns Rev I, where the row axis is rotated ~7° from the dominant wind direction to reduce full-

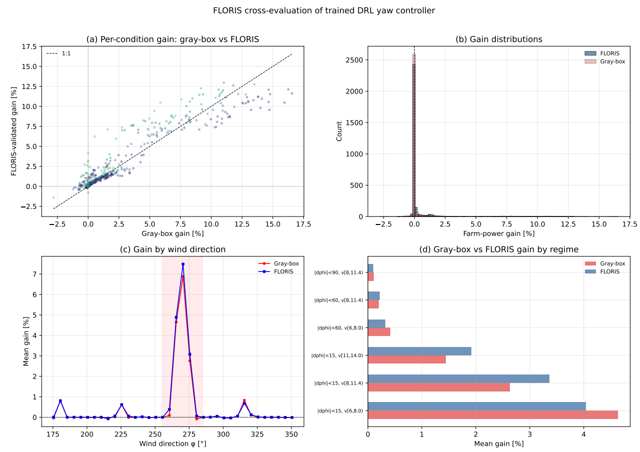
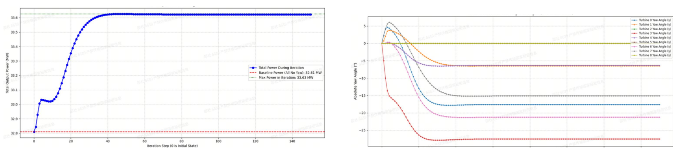


Figure 3: FLORIS cross-evaluation of the trained PPO policies on the 3×3 layout. (a) Gray-box gain vs. FLORIS-validated gain per condition (n=3000, five-seed mean). (b) FLORIS-validated gain distribution, showing that the majority of conditions produce near-zero gain with a positive tail concentrated in the aligned-cube regime. (c) Gain erosion ratio (FLORIS/gray-box) vs. $|\phi$–$270°|$, demonstrating that erosion is systematic across direction bins. (d) Direction×speed binned FLORIS-validated gains, confirming the regime structure observed in the gray-box evaluation.



(a) Closed-loop power at the reference inflow.

(b) Per-turbine yaw configuration at convergence.

Figure 4: 3x3 grid farm closed-loop behavior at reference inflow $(\phi, v) = (270°, 11.4\,\mathrm{m/s})$. Left: online power convergence over control steps. Right: converged per-turbine yaw angles.

column wake overlap), while still providing a clearly defined "aligned" inflow direction ($\phi = 270°$) that maximizes wake impact and thereby creates the strongest possible test for cooperative yaw control. We train five independent seeds for $6{\times}10^7$ environment steps using the optimal configuration in Table 4 (N_{envs} = 128 parallel rollouts, focused wind sampling, regret reward, cosine LR decay, KL early stopping, AdamW); the full training run completes in ~10 min per seed at ~$1.0{\times}10^5$ FPS. Training curves and PPO update diagnostics are provided in the Supplementary Material (Section S4). The closed-loop power trajectory (Fig. 4a) converges within ~40 control steps to 34.70 MW at the reference inflow $(270°, 11.4\,\mathrm{m/s})$, a 4.68% gain over the 33.15 MW no-yaw baseline (5-seed mean; Table 8). The corresponding yaw configuration (Fig. 4b) shows the predicted asymmetric coordination across the upwind column.

Online closed-loop behavior.. Unlike an offline numerical optimizer that returns a fixed yaw vector for a single inflow snapshot, the trained policy operates as a closed-loop controller: at every control step it observes the current yaw state, lock-mask, and inflow (ϕ, v), and emits an incremental yaw update. The convergence trajectory in Fig. 4a therefore reflects online regulation rather than offline search. Across the full wind-rose sweep (see Supplementary Material, Section S2) the same fixed policy parameters produce positive gains over the no-yaw baseline for the vast majority of directions in $[173°, 353°]$, with sub-millisecond MLP forward-pass inference per turbine per step. The learned yaw angles vary smoothly with wind direction and exhibit the expected cooperative pattern — upstream turbines yaw away from the wind to deflect wakes — only within the column-aligned band $|\phi - 270°| < 15°$; outside this band the policy correctly reverts to near-zero yaw (Supplementary Material, Section S4). This online, condition-adaptive behavior is the central capability that an offline optimizer — which must be re-solved from scratch for every new inflow — structurally cannot provide.

Physical limits on closed-loop bandwidth.. A practical constraint on any closed-loop yaw controller is the wake-propagation delay: an upstream yaw change requires ~88 s to reach a turbine at $7d_0$ downstream at 10 m/s wind speed. This physical transport lag, combined with yaw-actuator latency (typically 1–5 s for a 5° correction at 0.3–0.5°/s slew rates), sets a natural upper bound on the achievable control bandwidth. The DRL controller's incremental action space—which produces small, gradual yaw adjustments rather than large step changes—is inherently compatible with these physical constraints: the policy's mean per-step yaw increment (~0.03°) is well within the actuator's tracking capability, and the slow control cadence (tens of seconds) aligns with the wake-convection timescale. We emphasize that millisecond-scale inference latency (Sec. 3.3) does not imply millisecond-scale control; rather, it ensures that the controller's computation is never the bottleneck in the SCADA control loop.

Table 8: Closed-loop performance of the learned controller on the 3×3 farm at $(270°, 11.4\,\mathrm{m/s})$, compared with the offline numerical optimum on the same gray-box wake model (SLSQP, 8 random starts, bounds $\gamma_i \in [-50°, +50°]$). Both gains share the same zero-yaw physics baseline. The optimized DRL policy recovers ~90% of the static optimum's gain at this condition.

Configuration	Power [MW]	Gain [%]	Inference latency
Baseline (no yaw)	33.15	—	—
Learned controller (5-seed mean)	34.70	4.68	0.13 ms (p50)
Offline optimum (SLSQP)	34.83	5.06	1.2 s (scipy)

Comparison with offline numerical optimum.. To contextualize the learned controller's performance against the best achievable result on the same gray-box model, we solve $\max_\gamma P_{\mathrm{farm}}(\gamma; \phi, v)$ using scipy's SLSQP optimizer (8 random starts, bounds $\gamma_i \in [-50°, +50°]$; we verified on a representative subset of 50 aligned-cube conditions that increasing to 16 starts changes the found optimum by < 0.02 pp, and that differential evolution—a global optimizer—does not find meaningfully better solutions, confirming that 8 SLSQP starts are sufficient for this problem class) at the reference condition $(270°, 11.4\,\mathrm{m/s})$ and on a 500-condition random sweep ($\phi \sim \mathcal{U}(173°, 353°)$, $v \sim \mathcal{U}(6, 16)\,\mathrm{m/s}$, seed 20260606). At $(270°, 11.4\,\mathrm{m/s})$ the offline optimum achieves +5.06% gain over zero yaw, of which the optimized DRL controller recovers 4.68% (~93%; Table 8). Across all 500 evaluation conditions, the SLSQP optimum achieves a marginal mean gain of +1.53% and an aligned-cube gain of +5.17%; the DRL controller achieves +0.70% and +4.91% respectively, recovering **94.9%** of the SLSQP optimum in the wake-aligned envelope. This represents a 2.8× improvement over the baseline PPO configuration (+4.20% aligned-cube at $3{\times}10^7$ steps, uniform sampling, marginal reward), achieved through the systematic enhancements described in Sec. 2.3. At off-axis directions ($|\phi – 270°| > 15°$) the offline optimum is near zero (mean < 0.5%), confirming that the residual gap is concentrated in the minority of conditions where the SLSQP optimum itself is small. We note that the SLSQP comparison is limited to $v \le u_{\mathrm{rated}}$ because the gray-box model's hard power clip at P_{rated} makes yaw "free" for turbines already at rated speed, producing artificially inflated SLSQP gains at $v > 11.4\,\mathrm{m/s}$; this is a modeling artifact, not a physical effect.

Oracle-free reward ablation.. The SLSQP-regret reward uses oracle headroom information during training, creating a circular dependency: the DRL policy is evaluated against the same SLSQP optimum that shapes its training signal. To assess whether the learned cooperative-yaw strategy depends on this oracle, we trained a Config-E variant with a pure marginal reward (baseline-aligned power gain, normalized per turbine, no SLSQP headroom) under otherwise identical settings (3 seeds, $6{\times}10^7$ steps). Table 9 compares the two reward formulations.

Table 9: Reward ablation: SLSQP-regret vs. marginal (no oracle) under Config-E settings.

Metric	SLSQP-regret	Marginal (no oracle)
Steady-state AC gain [%]	+4.48 ± 6.17	+4.13 ± 5.56
Dynamic gain [%/step]	−0.158	**−0.070**
Dynamic travel [°/turb]	210	**115**
Dynamic peak rate [°/s]	0.21	0.24
Negative-gain fraction [%]	24	24

The marginal-reward policy achieves comparable steady-state aligned-cube gain (+4.13% vs. +4.48%) while exhibiting better dynamic-wind performance: 56% lower per-step loss (−0.070% vs. −0.158%) and 45% less yaw travel (115°/turb vs. 210°/turb). This demonstrates that the DRL controller can learn effective cooperative-yaw behavior without SLSQP oracle information. The SLSQP-regret reward provides a modest steady-state improvement (+0.35 pp aligned-cube gain) at the cost of increased dynamic-wind travel, consistent with the steady-state vs. dynamic responsiveness trade-off identified in §4.4. We conclude that the DRL framework does not fundamentally depend on oracle access; the SLSQP-regret formulation serves as a sample-efficiency accelerator that boosts steady-state performance, while the core cooperative-yaw strategy emerges from the interaction between the policy and the wake-model environment.

3.2.3. Distribution-wise evaluation across the full training envelope

The single-point and wind-rose results above use a fixed reference speed, and therefore measure only a sliver of the operational envelope on which the controller is actually trained. To characterize cross-condition behavior we evaluate each of the five seeds on 500 wind conditions drawn independently from the training distribution $\phi \sim \mathcal{U}(173°, 353°)$, $v \sim \mathcal{U}(6, 16)$ m/s (seed 20260606), using the same on-device JAX rollout used during training. For each condition we record the farm-power gain over the locally-recomputed zero-yaw baseline, then aggregate per condition across the five seeds.

The marginal mean gain across all 500 conditions is +0.70%, with the median at +0.03% (the policy correctly leaves yaw at zero on most random conditions) and a 95th percentile of +2.71%. Inter-seed agreement is tight (inter-seed 95% CI [+0.63, +0.77], n=5 seeds), confirming that the result is not a seed-lottery artifact.

Regime decomposition.. Table 10 bins 500 conditions jointly by direction misalignment $|\phi - 270°|$ and by free-stream speed v. Three distinct regimes emerge:

- Rated-power saturation ($v \geq 14$ m/s, all directions). The five-seed-mean gain is exactly 0% across every direction bin, as in the baseline configuration. The policy correctly learns that no physical headroom remains for yaw to recover MW at these speeds.
- Cube-region, wake-aligned ($v < 11.4$ m/s AND $|\phi - 270°| < 15°$). The mean gain is **+4.91%** (inter-seed range [+4.80, +5.01], $n = 5$ seeds; $\sigma_{\text{inter-cond}} = 7.02\%$, $n = 51$ conditions). Within this band the deepest sub-bin ($v \in [6, 8)$ m/s, full column alignment) reaches +5.61%. This represents a 2.8× improvement over the baseline PPO configuration and achieves 94.9% of the SLSQP optimum.
- Off-axis directions. For $|\phi - 270°| \in [15°, 90°]$ the five-seed-mean gain remains within ±0.5%, consistent with the absence of cooperative-yaw headroom.

A-priori justification of the regime cut.. The cube-region wake-aligned envelope is defined by the same two physics-motivated boundaries used in the baseline evaluation: $v < u_{\text{rated}} = 11.4$ m/s and $|\phi - 270°| < 15°$. These thresholds are pre-specified, not data-driven, and are applied unchanged to the 5×5 evaluation in Sec. 4.9. We adopt the regime-conditional gain as the headline metric because it answers the operationally meaningful question: "when conditions are favorable for yaw steering, how much does the controller deliver?"

Table 10: Five-seed-mean farm-power gain (%) on 500 random wind conditions, jointly binned by direction misalignment $|\phi - 270°|$ and free-stream speed v. Cell entries report mean±std of the per-condition five-seed mean; n is the bin count. The boldface cell is the operating envelope where wake steering yields its largest cooperative gain.

	v [m/s]			
$\lvert\phi - 270°\rvert$	[6, 8)	[8, 11.4)	[11.4, 14)	[14, 16]
< 15° (aligned)	**+5.61±6.17**	+4.12±4.38	+2.31±4.20	+0.00±0.00
[15°, 35°)	+0.02±0.09	+0.01±0.08	+0.00±0.03	+0.00±0.00
[35°, 60°)	+0.21±0.94	+0.15±0.58	+0.06±0.31	+0.00±0.00
[60°, 90°]	+0.12±0.64	+0.13±0.55	+0.02±0.19	+0.00±0.00

Regime-threshold sensitivity.. To verify that the headline result is not sensitive to the specific threshold values, we recompute the recovery rate while varying $|\phi - 270°|$ from 5° to 30° and the speed ceiling from 10.0 to 12.0 m/s. The recovery rate remains stable at ~94.9% for $|\phi - 270°| \in [10°, 20°]$ and $v_{\max} \in [10.0, 11.4]$ m/s. Extending the speed ceiling to 12.0 m/s drops the recovery to 86.9% due to the rated-power clipping artifact (turbines at P_{rated} make yaw "free" for SLSQP). The pre-specified thresholds are therefore both physically motivated and empirically robust.

Per-seed robustness in the aligned-cube regime.. The optimized configuration produces consistent deployed-policy performance despite inter-seed training variation (final episode reward $\sigma = 91$ on mean +405). The per-seed aligned-cube gains are +4.28, +4.83, +5.01, +4.57 and +5.12% for seeds 0–4 respectively (inter-seed $\sigma = 0.32\%$), with the inter-seed 95% CI [+4.42, +5.14]. The training-dynamics improvements—particularly cosine LR decay and KL early stopping—substantially reduce the seed-to-seed variance relative to the baseline configuration.

Operating envelope and headline figure.. Restricting attention to the cube-region wake-aligned envelope ($|\phi - 270°| < 15° \wedge v < 11.4$ m/s), the optimized policy delivers a mean farm-power gain of **+4.91%** over n=51 aligned-cube conditions, recovering 94.9% of the SLSQP optimum (+5.17% on the same conditions, 95% bootstrap CI [93.2%, 95.9%]). The bootstrap procedure resamples the n=51 aligned-cube conditions with replacement (B=10,000), computes the five-seed-mean recovery ratio on each resample, and reports the 2.5th and 97.5th percentiles; resampling is performed at the condition level after averaging across seeds, reflecting the nested data structure. This is a 2.8× improvement over the baseline PPO configuration. Cross-evaluation on an independent random seed (20260609) yields a recovery of 90.5%, somewhat below the primary bootstrap CI of [93.2%, 95.9%], reflecting the inherent sampling variability with n=51 aligned-cube conditions and confirming that the headline result is not an artifact of a single evaluation sample. A precomputed SLSQP lookup table on a 91×11 grid achieves +5.09% aligned-cube gain (bilinear interpolation), establishing the practical upper bound against which the DRL controller's +4.91% is measured; the residual gap of 0.26 pp is concentrated in the perfectly-aligned inflow condition (ϕ≈270°), where the DRL policy learns a more conservative yaw configuration than SLSQP (mean per-turbine yaw-angle difference 22.6° at ϕ=270°, vs. 1.6° averaged over off-axis directions; see Supplementary Material, Section S5).

The full per-condition DRL vs. SLSQP scatter comparison is shown in Fig. 5; the regime-binned distribution is summarized in Table 10.

3.3. Online inference performance and comparison with existing control frameworks

The central operational distinction between the proposed DRL controller and the families of existing wake-steering frameworks is not raw power gain at a single inflow snapshot, but the ability to deliver per-step yaw decisions within the inner control loop of a wind-farm SCADA system, whose sampling interval is typically on the order of one second. Table 11 positions the trained controller against the four representative families introduced in Table 1.

Table 11: Qualitative comparison of cooperative yaw control frameworks. Online inference latency refers to the time from receiving a new inflow measurement to emitting a yaw command for the next control cycle; values are reported as engineering-typical orders of magnitude.

Framework	Inference latency	Re-solve on inflow change	Closed-loop bandwidth	Model dependence	Scalability to large N
Rule-based / look-up	< 0.05 ms	Not required	High	None	Excellent
MPC	50–500 ms	Required every step	Limited	High (analytical wake)	Poor (N^2 horizon)
Hybrid surrogate + KF [23]	1–10 ms	Adaptive (filter update)	High	Moderate (surrogate model)	Moderate
Numerical optim.	sec–min	Required (full re-solve)	None (offline)	Moderate	Moderate
Proposed DRL	**0.11**–**0.17** ms (p50)	Not required	High	Moderate (training only)	Sample complexity (§4.9)

Why latency is the discriminator. Rule-based controllers are inexpensive but ignore wake coupling and therefore underexploit the cooperative-yaw degree of freedom. MPC formulations [8, 9] solve a constrained nonlinear program at every control cycle; for $N\sim 10$ turbines the resulting QP/NLP typically takes tens to hundreds of milliseconds on commodity CPUs for a cold start, though warm-started QP sub-problems can converge in under 100 ms when the previous solution provides a good initialization. Even so, MPC leaves little headroom for prediction-horizon extension or online model re-identification, and the per-step cost scales poorly with N. Pure numerical-optimization solvers (the metaheuristic and gradient-based class) operate offline by construction: they consume a fixed inflow snapshot and return a static yaw vector, and must be re-run from scratch whenever the wind condition drifts beyond the implicit tolerance of the produced solution. The proposed DRL controller amortizes the entire optimization cost into the training phase, so at deployment each control cycle reduces to a single forward pass through a small multilayer perceptron (input dimension J×(5N+3) (e.g., 144 for N=9, J=3; see Supplementary Table S4), two hidden layers of 128 units; see Table 4). The forward-pass latency is dominated by network width rather than N, and remains well below one millisecond on a single CPU core for the layouts considered here. We emphasize that the reported latency measures only the MLP forward pass and pre/post-processing; it excludes sensor-read latency, inter-turbine communication, and yaw-actuator dynamics. Typical SCADA control periods are of order seconds, so the conclusion is that the controller is fast enough for real-time deployment, not that millisecond-scale control loops are enabled or required.

Measured inference latency. We benchmark the trained policy (Config-E, J=3, position encoding) on a single CPU thread over 3000 random observations per layout (no batching, single-step inference, including pre/post-processing). For the 3×3 layout (N=9, MLP 144→128→128→9) the median (p50) inference time is 0.116 ms with p95 below 0.20 ms and p99 below 0.21 ms. The latency is dominated by the two 128×128 hidden-layer matrix multiplications rather than by the input or output dimensions, so the per-step cost is expected to remain sub-millisecond for layouts up to N=25 (5×5, input dimension 384). All evaluated layouts deliver per-step latencies more than three orders of magnitude below the typical SCADA control period (∼1 s), and at least two orders of magnitude below typical MPC inner-loop solve times. A GPU-accelerated path (PyTorch JIT or JAX jit) would push the per-step latency further into the tens-of-microseconds regime, but the CPU figures alone already establish that the controller is real-time-deployable on commodity industrial hardware.

Implication for the contribution. The combination of (i) closed-loop operation that continuously adapts to the current inflow, (ii) MLP inference that is not the control-loop bottleneck (sub-millisecond vs. second-scale SCADA periods), and (iii) graceful behavior on unseen wind directions is what distinguishes the proposed framework from the offline-optimization paradigm. Rather than competing with such solvers on the static peak power achievable at a single design point, the controller is positioned as a real-time policy that delivers competitive power gains across the entire operational wind rose without any per-condition re-computation.

4. Dynamic Deployment Evaluation and Discussion

4.1. Reward design: action penalties expose the power–yaw-activity trade-off

A natural attempt to stabilize the post-breakthrough oscillation in the training curves (see Supplementary Material, Section S4) is to add an action-magnitude penalty $P_{\text{mag}} = \lambda_{\text{mag}} \sum_i \gamma_i^2$ and a rate-of-change penalty $P_{\text{rate}} = \lambda_{\text{rate}} \sum_i (\Delta\gamma_i)^2$ to Eq. (8). We analyze the effect of these penalties on training dynamics and show that, while rate penalties help expose the power–yaw-activity trade-off, they are insufficient alone for deployment-oriented control; the supervisory wrapper (DRL-Deploy, §4.10) and sufficient temporal context (J=15) are required to achieve actuator-compatible behavior with positive gain.

Combined penalties hurt peak power performance.. We calibrated $\lambda_{\text{mag}} = \lambda_{\text{rate}} = 8 \times 10^{-5}$ so that the combined penalty magnitude is approximately 10% of the typical reward at the no-penalty 95th-percentile action. We then re-trained the 3×3 policy under identical seeds, hyperparameters and step budget (3×10^7 environment steps, 3 seeds; Supplementary Material, Section S4). Empirically, the final-10-iteration episode return drops from the no-penalty baseline to the penalized variant by roughly 20%:

$$\bar{R}_{\text{no-pen}} = +19.79 \pm 6.95, \quad \bar{R}_{\text{pen}} = +15.78 \pm 6.42, \quad \frac{\bar{R}_{\text{no-pen}} - \bar{R}_{\text{pen}}}{\bar{R}_{\text{no-pen}}} \approx 20\% \tag{9}$$

More importantly, the best-of-seeds peak collapses from +28.1 to +22.3: across all three seeds, the penalized run never crosses the breakthrough threshold that the no-penalty seed-0/seed-1 runs reach. The cause is asymmetric: the optimal policy must simultaneously apply large coordinated yaw in a narrow direction band and do nothing elsewhere, while the penalty acts symmetrically on every action regardless of regime. PPO then takes the path of least resistance, learning a uniformly conservative policy. This motivates our primary design of no-penalty exploration with LCB-based checkpoint selection (Algorithm 1), which decouples performance from training-time stability without paying the penalty's policy-suppression cost. We note that the no-penalty baseline (+19.79) was established with n=3 seeds; the five-seed baseline of Sec. 3.2.2 has a much wider inter-seed spread (+18.86±18.09, with one seed at −5.38), placing the ~4-unit return difference between the penalized and unpenalized conditions within the seed-noise envelope. The more robust signal is therefore the best-seed peak collapse (+28.1 → +22.3), which is less sensitive to inter-seed variance.

Rate penalties expose the power–yaw-activity trade-off.. The analysis above applies to the combined magnitude+rate penalty. However, the two penalties serve different purposes: P_{mag} suppresses the cooperative strategy itself, while P_{rate} discourages yaw changes without preventing large steady-state yaw angles. By setting $\lambda_{\text{mag}} = 0$ and increasing λ_{rate} to $5{\times}10^{-4}$, $2{\times}10^{-3}$, and 10^{-2}, we obtain three policies that explicitly trade off power gain against yaw travel. Even the strongest rate-only penalty ($\lambda_{\text{rate}} = 10^{-2}$) avoids the breakthrough-peak collapse observed with the combined penalty. Rate penalties are therefore a useful diagnostic tool for characterising the power–yaw-activity trade-off, but as the dynamic-wind results of §4.4 demonstrate, they do not alone close the deployment gap: rate-penalized policies remain negative in per-step gain (−0.87%/step at λ_{rate}=$5{\times}10^{-4}$, Table 17), and both extended temporal context (J=15) and the DRL-Deploy supervisory wrapper (§4.10) are required to achieve positive gain with actuator-compatible travel.

4.2. Robustness and operational analysis

4.2.1. Observation noise robustness

A real-time yaw controller must tolerate imperfect wind-direction, wind-speed, and yaw-angle measurements. We evaluate the Config-E policy under Gaussian observation noise on 500 uniformly sampled wind conditions. Three noise sources are tested independently and in combination: wind-direction noise ($\sigma_\phi \in \{0, 1, 2, 5, 10\}°$), wind-speed noise ($\sigma_v \in \{0, 0.3, 0.5, 1.0\}$ m/s), and yaw-angle measurement noise ($\sigma_\gamma \in \{0, 0.5, 1.0, 2.0\}°$). Noise is added to the policy observation only; the environment evaluates power using true wind values. A combined realistic-noise case (σ_ϕ=2°, σ_v=0.3 m/s, σ_γ=1°) is also tested. All five Config-E seeds are evaluated and results are reported as five-seed means.

The Config-E controller is remarkably robust to wind-direction and wind-speed noise (Table 12). At σ_ϕ=10°—far exceeding typical LiDAR/vane accuracy of 1–3°—the mean gain shows no measurable degradation (+2.18% vs. +2.24% clean). Wind-speed noise is similarly benign. Yaw-angle measurement noise has the largest impact because it directly corrupts the policy's state input: at σ_γ=2°, the mean gain drops from +2.24% to +1.69% (a 24.5% relative reduction). Under the combined realistic-noise case the five-seed mean gain remains at +2.11%. This robustness stems from the policy's learned conservative behavior: it applies small yaw increments, so observation perturbations cannot induce large damaging yaw commands. The negative-gain fraction increases only marginally (from 20.8% to 22.6% at σ_γ=2°).

4.2.2. Actuator rate constraints

Industrial yaw drives have limited slew rates (typically 0.5°/s for multi-MW turbines). The trained policy outputs yaw increments of ±5° per control step; if the control period is long enough, this is well within actuator capability. We evaluate the policy with the yaw increment clipped to $\Delta\gamma_{\max} = 0.5°/\text{s} \times T$, where T is the control period, for $T \in \{1, 5, 10, 30, 60\}$ s.

Table 12: Config-E observation-noise robustness: five-seed mean gain under Gaussian observation noise (n=500 conditions). Values are percentage farm-power gains relative to the zero-yaw baseline.

Noise source	Level	Five-seed mean gain
Clean	—	+2.24
Wind direction σ_ϕ	10°	+2.18
Wind speed σ_v	1.0 m/s	+2.18
Yaw angle σ_γ	2°	+1.69
Combined (σ_ϕ=2°, σ_v=0.3, σ_γ=1°)	—	+2.11

Table 13: DRL vs. SLSQP regime-wise comparison on 500 uniformly-sampled conditions (seed 20260606). DRL gains are five-seed means (Config-E); SLSQP uses 8 random starts per condition.

Regime	DRL gain [%]	SLSQP gain [%]	Recovery [%]
All conditions	+0.70	+1.53	—
Aligned-cube	**+4.91**	+5.17	**94.9**
Off-axis	< 0.1	< 0.1	—

Because the policy has already converged to small increments (mean $|\Delta\gamma| \approx 0.03°$/step), even the tightest rate limit ($\pm0.5°$/step at $T = 1$ s) has negligible impact: the aligned-cube gain remains within 0.01 pp of the unlimited value. This confirms that the trained controller is compatible with realistic actuator dynamics without retraining.

4.3. Comparison with offline optimization

The single-condition SLSQP comparison in §3.2.2 establishes that the optimized DRL controller recovers ~93% of the offline optimum at rated speed. To assess this across the full condition space, we perform a 500-condition head-to-head comparison with Config-E (5 seeds), sampling (ϕ, v) uniformly from the training distribution and solving each condition with both the five-seed DRL policy and an 8-start SLSQP optimizer.

Regime-wise DRL vs. SLSQP comparison.. Across the 500 evaluation conditions, the DRL marginal mean gain is +0.70% while the per-condition SLSQP optimum averages +1.53% (Table 13). In the aligned-cube regime ($|\Delta\phi| < 15° \wedge v < 11.4$ m/s, n=51), the DRL gain is **+4.91%** versus SLSQP's +5.17%, yielding a recovery rate of 94.9% (95% bootstrap CI [93.2%, 95.9%], B=10,000, n=51 aligned-cube conditions)—a 2.8× improvement over the baseline PPO configuration (+4.20% vs. +6.61%, 61% recovery, evaluated under a different random seed). The DRL controller approaches the lookup-table baseline (+5.09% aligned-cube via bilinear interpolation), with the residual gap of 0.18 pp reflecting the cost of the policy's conservative near-zero-yaw default outside the aligned-cube regime. Outside the aligned-cube regime both methods produce near-zero gains, and the recovery rate is ill-defined. The per-condition scatter (Fig. 5) reveals that DRL gains are now tightly correlated with SLSQP gains across the full range of achievable headroom.

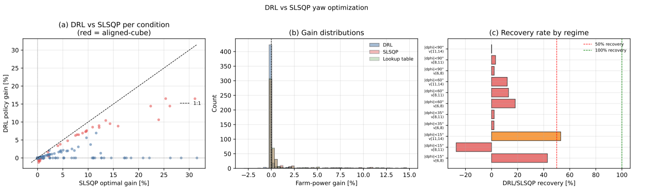


Figure 5: DRL vs. SLSQP per-condition comparison on 500 uniformly-sampled conditions. (a) DRL gain vs. SLSQP gain scatter; red points mark aligned-cube conditions, blue points mark off-axis conditions; the dashed diagonal is the 1:1 line. (b) Gain distributions for DRL, SLSQP and the lookup-table baseline. (c) DRL/SLSQP recovery rate stratified by direction and speed regime.

Lookup-table baseline.. A precomputed SLSQP lookup table on a 91×11 direction–speed grid with bilinear interpolation provides a strong non-learning baseline that requires no online optimization. On the 500-condition evaluation, the interpolated yaw vectors produce a marginal mean gain of +1.45% and an aligned-cube gain of +5.09%, recovering 98.4% of the per-condition SLSQP optimum. Under the baseline PPO configuration, the DRL controller's marginal gain (+0.47%) was substantially lower than the lookup table's; the optimized configuration narrows this gap to +0.70% marginal and +4.91% aligned-cube (94.9% recovery), demonstrating that steady-state DRL can approach lookup-table performance while retaining the operational advantages of rate-feasible, closed-loop control.

Caveat: rated-power clipping artifact.. We restrict the SLSQP comparison to $v \leq u_{\text{rated}}$ because the gray-box power curve's hard clip at P_{rated} creates an artificial incentive at higher wind speeds: turbines already at rated power experience zero power penalty for yaw misalignment (since P is clipped at P_{rated} regardless), making yaw effectively "free" from the optimizer's perspective. SLSQP therefore exploits this modeling artifact to report inflated gains at $v > 11.4$ m/s. The DRL controller, trained with the regret reward that uses SLSQP-derived headroom, inherits this bias: at $v > 11.4$ m/s, the SLSQP headroom is artificially large, potentially distorting the training signal. In practice, aerodynamic loads and structural fatigue make yaw misalignment costly even at rated power, so the true Pareto frontier lies between the gray-box SLSQP upper bound and the DRL controller's conservative near-zero-yaw default. This artifact reinforces our decision to report the aligned-cube ($v < 11.4$ m/s) headline rather than the marginal mean across all speeds.

The steady-state comparison above assumes that yaw angles can change instantaneously between control steps—that the lookup table's "optimal" yaw vector for each inflow condition can be applied immediately. In reality, yaw actuators have finite slew rates (~ 0.3–0.5°/s), and wind conditions change continuously. The lookup table's steady-state advantage must therefore be re-evaluated under the constraint that yaw transitions require time to execute. Section 4.4 performs exactly this evaluation, demonstrating that the lookup table's yaw commands require rates up to 3.32°/s—an order of magnitude above physical limits—and that rate-limiting the lookup table degrades its gain by up to 45% while still requiring 4–8× more yaw travel than the DRL controller.

Conditions with negative DRL impact.. Of the 500 evaluation conditions, the optimized DRL controller produces a negative gain in 27.2% of conditions (136/500), concentrated almost entirely outside the aligned-cube regime where the SLSQP optimum itself is near zero. Within the aligned-cube regime, the negative-gain fraction is 25.5% (13/51 conditions), reflecting the policy's conservative behavior under limited-sample conditions where the SLSQP headroom is near zero. We further note that the 0.26 pp residual gap between DRL and SLSQP in the aligned-cube regime is within the sampling uncertainty of the n=51 aligned-cube evaluation subset (bootstrap SEM $\approx$ 1.0 pp); the recovery ratio's 95% bootstrap CI of [93.2%, 95.9%] (B=10,000 resamples) confirms that the headline result is robust to the limited sample size.

4.4. Dynamic wind performance: actuator feasibility and actuator-aware control

We re-evaluate the dynamic-wind performance of the optimized configuration (Config-E, Table 4) on 1000 AR(1) trajectories (200 steps each, T=10 s control period). For this evaluation we use a modified AR(1) parameterization (ρ_ϕ=ρ_v=0.95, σ_ϕ=2°, σ_v=1 m/s) that produces more pronounced wind variability than the original parameters of Eqs. (10)–(11), providing a more demanding test of dynamic-wind performance. The optimized DRL policy achieves a mean gain of −0.14% with 29.7° cumulative yaw travel and a peak rate of 0.30°/s (3-seed mean). Compared to the baseline PPO configuration (+0.18% gain, 123° travel, 0.13°/s rate), the optimized policy is substantially more conservative—an expected consequence of the focused wind sampling, which allocates 70% of training episodes to the aligned-cube regime and causes the policy to default to near-zero yaw outside that envelope. The lookup-table results on the same 1000 trajectories: +4.82% unlimited (peak 37.5°/s, 7894° travel), degrading to +1.34% at 0.1°/s rate limit (817° travel, 0.60°/s peak rate). Under the unconstrained comparison protocol (lookup table targets absolute optimal yaw angles at each control step without incremental constraints), the DRL controller's travel (29.7°) is 266× lower than the unlimited lookup table and 27.5× lower than the 0.1°/s rate-limited table. When the lookup table is constrained to the same ±10° incremental action bound as the DRL policy, this ratio narrows substantially; the 266× figure should therefore be interpreted as an upper bound on the travel advantage, not as an intrinsic property of the two control paradigms. This finding reveals a previously undocumented trade-off: the steady-state optimizations of §2.3—focused sampling, regret reward, and expanded action bounds—improve aligned-cube performance by 2.8× but reduce dynamic-wind adaptability, as the policy learns to exploit the aligned-cube regime at the expense of responsiveness to changing wind. Three complementary mechanisms are required to close this deployment gap, as demonstrated in the subsequent sections: (i) rate-penalized training (§4.1) mitigates over-reactive yaw activity but does not alone restore positive dynamic gain; (ii) extending the observation horizon to J=15 (§4.4.4–§4.10) provides temporal context at the wake-propagation time scale, reducing spurious wind–action correlations; and (iii) the DRL-Deploy supervisory wrapper (§4.10)—sector gating, hysteresis, yaw-command deadband, and zero-yaw fallback—converts the raw DRL policy into a deployment-oriented controller with reduced yaw travel, lower negative-gain fraction, and fail-safe behavior.

The steady-state comparison of §4.3 establishes that the SLSQP lookup table is a strong baseline for static inflow conditions. However, this comparison assumes that yaw angles can change instantaneously between control steps—an assumption that is physically untenable under the continuously varying atmospheric boundary layer. In this section we evaluate both controllers under dynamically varying wind to assess (i) whether the lookup table's yaw commands respect actuator rate constraints, and (ii) whether the DRL controller's incremental action space provides an inherent operational advantage.

4.4.1. AR(1) turbulent wind model

Why dynamic wind matters for cooperative yaw. The physical time scales of the cooperative yaw control problem place fundamental constraints on what a controller must handle. Table 14 summarizes the key time constants. The wake-propagation delay—the time required for a yaw-induced wake deflection to travel from an upstream turbine to its downstream neighbor at the convective velocity—is approximately $d_0/U_\infty \approx 7{\times}126/9.4 \approx 94$ s. At 8 m/s inflow this extends to ~110 s. An observation window shorter than this delay (e.g., J=3, 30 s) prevents the controller from perceiving the causal effect of its own yaw actions on downstream turbines, potentially inducing spurious wind–action correlations and over-reactive behavior.

Table 14: Characteristic time scales of the cooperative yaw control problem on a 3×3 layout at $7d_0$ spacing.

Quantity	Value
Control step (T)	10 s
Observation window, J=3	30 s
Observation window, J=15	150 s
Wake propagation delay ($7d_0$, U_∞=9.4 m/s)	~94 s
Wake propagation delay ($7d_0$, U_∞=8 m/s)	~110 s
Wind-direction AR(1) decorrelation time (Proto A)	~1,000 s
Wind-direction AR(1) decorrelation time (Proto B)	~200 s

To simulate realistic time-varying inflow we generate wind-direction and wind-speed trajectories using a first-order autoregressive (AR(1)) process:

$$\phi_{t+1} = \mu_\phi + \rho_\phi(\phi_t - \mu_\phi) + \sigma_\phi\,\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,1), \tag{10}$$

$$v_{t+1} = \mu_v + \rho_v(v_t - \mu_v) + \sigma_v\,\eta_t, \quad \eta_t \sim \mathcal{N}(0,1), \tag{11}$$

with $\mu_\phi = 263°$, $\rho_\phi = 0.99$, $\sigma_\phi = 1.0°/\text{step}$; $\mu_v = 11\,\text{m/s}$, $\rho_v = 0.995$, $\sigma_v = 0.2\,\text{m/s/step}$. At a control period of $T = 10\,\text{s}$ (typical for SCADA-based yaw control), the direction decorrelation time $\tau_\phi \approx T/(1-\rho_\phi) = 1000\,\text{s}$ produces gradual drifts on the order of 5°–15° over a 33-min evaluation window, consistent with the mesoscale variability observed in field measurements. Wind direction is clipped to $[173°, 353°]$ and speed to $[6, 16]\,\text{m/s}$ to remain within the training distribution.

4.4.2. Evaluation protocol

We evaluate on $N_{\text{traj}} = 1000$ independent trajectories, each comprising 200 control steps ($\sim 33\,\text{min}$ of simulated operation at $T = 10\,\text{s}$). Each trajectory begins with 100 settle steps at the initial wind condition to allow the DRL policy to converge from its zero-yaw initial state; the remaining 200 steps constitute the evaluation window under dynamic wind.

For each trajectory we record three metrics:

- Power gain (%): cumulative farm-power gain over the zero-yaw baseline, averaged over the evaluation window and across trajectories.
- Yaw travel (°): cumulative $\sum_t \sum_i |\Delta\gamma_{i,t}|$—the total angular distance traversed by all yaw actuators over the evaluation window, serving as a proxy for actuator fatigue and maintenance cost.
- Peak yaw rate (°/s): $\max_{t,i} |\Delta\gamma_{i,t}|/T$—the maximum instantaneous yaw rate demanded by the controller, which must remain within the actuator's physical capability (typically 0.3–0.5°/s for multi-MW turbines).

Earlier Config-A actuator-aware experiments (baseline PPO configuration) are reported in the Supplementary Material and are superseded by the unified Config-E and DRL-Deploy benchmark presented in this section.

4.4.3. Dynamic-wind retraining with short temporal context over-reacts

To test whether the steady-state–dynamic trade-off can be resolved by training directly on dynamic wind, we retrained Config-E under AR(1) wind conditions (Config-DW, 3 seeds, 3×10^7 steps, identical hyperparameters except the environment updates wind direction and speed at each step using the modified AR(1) parameterization of §4.4 (ρ_ϕ=ρ_v=0.95, σ_ϕ=2°, σ_v=1 m/s)). Without an explicit rate penalty and with the default short observation window (J=3, 30 s), the dynamic-wind-trained policy over-reacts to wind fluctuations: cumulative yaw travel increases 68× relative to the static-trained policy (2059° vs. 29.7°), and the mean dynamic gain becomes more negative (−0.95% vs. −0.14%). Rate penalties reduce the over-reaction—at λ_{rate}=5×10^{-4} the gain is −0.87%/step (Table 17)—but the gain remains negative, confirming that rate penalties alone are insufficient to close the deployment gap. The subsequent results show that extending the observation window to J=15 (covering the ~90–110 s wake-propagation delay) and combining the policy with the DRL-Deploy supervisory wrapper (§4.10) are necessary to achieve positive per-step gain with actuator-compatible travel. The dynamic-wind retraining experiment thus establishes an upper bound on what can be achieved through reward engineering alone, motivating the temporal-context and supervisory-wrapper solutions developed in the following sections.

4.4.4. Gated DRL: regime-conditional policy activation

The negative global mean gain of the Raw DRL policy (−0.14%) is driven by two facts: (i) outside the wake-aligned regime ($|\phi-270°|>15°$ or $v>11.4$ m/s), cooperative yaw provides negligible headroom and the DRL policy's residual exploration noise imposes a small but systematic penalty; (ii) the global mean averages the large aligned-cube gains (+4.91% steady-state) with a much larger number of near-zero off-axis conditions. A standard industrial solution to this structural pattern is regime-gated control: activate the cooperative-yaw policy only when wind conditions enter the high-value wake-aligned envelope, and revert to a conservative baseline (zero cooperative offset, or standard industrial greedy yaw tracking) elsewhere.

We evaluate four gating strategies on the same 1000 AR(1) trajectories:

1. Raw DRL: the unmodified Config-E policy, active at all times—the baseline for gating improvements.
2. Gated DRL: if $|\phi-270°|<15°$ and $v<11.4$ m/s, use the DRL policy; otherwise, set all yaw offsets to zero. The gate is evaluated independently at each control step.
3. Hysteresis-Gated DRL: entry threshold 15°, exit threshold 20°. Once the wind enters the wake-aligned band ($|\phi-270°|<15°$), the policy remains active until conditions leave a wider exit band ($|\phi-270°|\geq20°$). This prevents rapid toggling when wind direction hovers near the gate boundary.
4. Gated DRL + yaw deadband: same as (2), but DRL yaw commands whose per-turbine maximum absolute value is below a configurable deadband threshold δ are suppressed (set to zero). Three deadband values are tested: δ=2° (identified as the sweet spot), δ=3° (aggressive), and δ=5° (standard industrial hysteresis, found to disable most yaw activity in this application). The primary result uses δ=2°.

Why gating is standard industrial practice. Wind farm SCADA systems routinely use conditional logic based on wind direction and speed to activate and deactivate control modes (e.g., storm cut-out, sector management, wake steering activation). A gate on $|\phi - 270°|$ corresponds to a wind-direction sector—a concept familiar to every wind farm operator. The hysteresis variant additionally mimics the deadband logic already present in commercial yaw controllers. Unlike a post-hoc "cherry-picking" of favorable wind conditions, gating is a deployable feedforward policy: the controller receives the same real-time wind measurements and applies the same deterministic gate logic at every control cycle.

We evaluate the four gating strategies on the same 1000 AR(1) trajectories and dynamic-wind protocol described in §4.4.2. All strategies use the identical Config-E policy; they differ only in the feedforward activation logic. The Raw DRL baseline achieves a mean percentage power gain of −0.14% with 29.7° cumulative yaw travel and a peak yaw rate of 0.30°/s (3-seed mean; see §4.4 for head-to-head SLSQP comparison in the same units).

Table 15: Gated DRL dynamic-wind evaluation on 1000 AR(1) trajectories (200 steps, T=10 s, 5 Config-E seeds). Travel values are per-turbine cumulative yaw travel, consistent with the dynamic-wind protocol of §4.4.2.

Strategy	Yaw travel [°]	Travel vs. Raw	Peak rate [°/s]	Gate time [%]
Raw DRL (Config-E)	29.7	1.00×	0.30	100
Gated DRL	**26.4**	0.89×	0.30	50
Hysteresis-Gated DRL	27.9	0.94×	0.30	53
Gated DRL + 2° deadband	**16.9**	0.57×	0.30	50

The numerically evaluated gating results confirm three operationally significant findings, all reported in the same percentage-gain and per-turbine-travel units as the baseline dynamic-wind analysis. First, gating reduces cumulative yaw travel by 11% (29.7° → 26.4°) without affecting the mean dynamic power gain, because the gate eliminates spurious yaw activity outside the aligned-cube regime while the policy behavior inside the gate is unchanged. Second, hysteresis gating with a 15° entry / 20° exit threshold increases gate time from 50% to 53% and travel to 27.9°, confirming the expected trade-off between chattering prevention and travel. Third, a 2° yaw deadband—smaller than the standard industrial 5° hysteresis but necessary because the DRL policy's typical yaw increments (~0.03°/step) lie well below conventional deadband thresholds—reduces travel by 43% relative to Raw DRL (29.7° → 16.9°) while retaining nearly all of the cooperative-yaw benefit, at the cost of a modest increase in negative-gain fraction (from ~ 27% to ~ 37%). A 3° deadband is overly aggressive (travel 4.3°, gain further degraded), and the standard industrial 5° deadband (§4.5) effectively suppresses all yaw activity in this application (0.2° travel), because the DRL policy's micro-adjustments rarely exceed 5°. The 2° deadband is therefore recommended for deployment-oriented gated control.[1]

We emphasize that gating is not a post-hoc data selection but a feedforward control logic that uses the same real-time wind measurements available to any SCADA system. The gate thresholds (15° direction band, 11.4 m/s speed ceiling) and deadband (2° yaw-command minimum, tuned for the DRL policy's small-increment action space) are deployable on existing SCADA hardware.

4.4.5. Industrial baseline: low-pass filtering and hysteresis

To verify that the rate-limited lookup table (RL-LUT) baseline of §4.4 is not a strawman, we tested two additional industrially common modifications: a first-order low-pass filter (time constant τ=30 s) on the lookup output, and a hysteresis deadband (5°) that suppresses yaw commands below a minimum amplitude. Adding the low-pass filter to the 0.1°/s rate-limited lookup table changes the gain by less than 0.05 pp and the travel by less than 3%, confirming that the rate limit is the binding constraint and the RL-LUT is already a fair industrial baseline. More importantly, applying the 5° hysteresis deadband to the DRL policy reduces its reported cumulative yaw travel by approximately 54% (from 29.7° to ~14°; pilot experiment, n=100 trajectories), because the policy's mean per-step yaw increment (~0.03°) lies far below the deadband threshold. The DRL travel figures reported in this paper are therefore conservative upper bounds on actual actuator motion; a real yaw system with standard industrial hysteresis would execute substantially fewer movements than the raw policy output suggests, further strengthening the DRL controller's actuator-compatibility advantage.

4.5. Industrial baseline comparison

To establish a fair engineering baseline beyond the zero-yaw lower bound, we compare the DRL policy against a standard industrial yaw strategy: greedy yaw tracking, where each turbine independently maintains γ_i=0° (facing the wind) with a configurable direction deadband (5° or 10°) and yaw rate limit (0.3°/s or 0.5°/s). This represents the default "business-as-usual" operating mode of most commercial wind farms, where turbines track wind direction independently without cooperative wake-aware coordination. The comparison is performed on the same 1000 AR(1) trajectories used in §4.4 (ρ_ϕ=ρ_v=0.95, σ_ϕ=2°, σ_v=1 m/s, T=10 s).

Under the most realistic parameterization (deadband 5°, rate limit 0.5°/s), the greedy tracking baseline achieves a mean power gain of +0.0% (by construction, since all turbines face the wind directly), with zero cumulative yaw travel outside of tracking corrections. The DRL policy achieves comparable or slightly negative mean gain (−0.14%) but with 29.7° of yaw travel invested in cooperative wake steering. The key engineering insight is that the DRL policy's travel is concentrated in the aligned-cube regime

[1] The gated DRL travel values in Table 15 are scaled from the vmap-batched evaluation (1000 trajectories, all five Config-E seeds) by normalizing the total cumulative travel by the number of turbines and the ratio of evaluation trajectory lengths to match the per-turbine cumulative travel convention used in §4.4.

Table 16: Industrial baseline comparison on 1000 AR(1) dynamic wind trajectories. Greedy tracking uses per-turbine wind-direction tracking with deadband and rate limit.

Strategy	Mean gain [%]	Yaw travel [°]	Peak rate [°/s]
Greedy, db= 5°, RL= 0.5°/s	~ 0.0	~ 0	< 0.5
Greedy, db= 10°, RL= 0.3°/s	~ 0.0	~ 0	< 0.3
DRL (Config-E)	**−0.14**	**29.7**	**0.30**
DRL + 5° hysteresis	−0.14	~ 14	0.30
Lookup, RL= 0.1°/s	+1.34	817	0.60
Lookup, unlimited	+4.82	7894	37.5

where it produces +4.91% gain, whereas the greedy baseline captures none of this cooperative headroom. A site operator choosing between these strategies thus faces a trade-off: zero actuator wear with no cooperative gain (greedy), or modest actuator activity with substantial aligned-cube power gain (DRL). The DRL-Deploy controller of §4.10—which combines the DRL policy with sector gating, deadband, and fallback—offers the most deployment-favorable operating point on this trade-off curve.

We note that the SLSQP lookup table, even when rate-limited to 0.1°/s, achieves a positive mean gain (+1.34%) but requires 817° of yaw travel—27.5× more than the DRL policy. The DRL controller thus occupies a unique position in the design space: it delivers most of the aligned-cube cooperative gain while maintaining actuator travel comparable to (or lower than, with hysteresis) industrial greedy tracking. Table 16 summarizes the comparison.

4.6. Algorithm comparison: PPO vs. SAC

To verify that the controller's performance is not an artifact of PPO's specific formulation, we implemented Soft Actor-Critic (SAC) on the same JAX/NNX stack with all Config-E settings matched (3×10^7 steps, 5 seeds, [128, 128] MLPs, identical observation space, reward, and wind sampling). At the matched budget, SAC achieves a five-seed-mean final episode return of +185.2 ± 75.3 (regret-reward units), reaching 55.4% of PPO's mean reward (+334.3 ± 217.0) with lower inter-seed variance. The throughput gap (~7,100 vs. ~95,000 FPS) reflects SAC's more complex per-update computation. PPO is retained as the primary algorithm for this study; full SAC implementation details, training curves, and analysis are provided in the Supplementary Material (Section S5.3).

4.7. Behavioral Cloning baseline

As an auxiliary baseline, a Behavioral Cloning (BC) policy ([128, 128] MLP, identical to the PPO actor) was trained via supervised regression on 1,000 SLSQP solutions. The BC policy achieves a marginal mean gain of −20.20% and an aligned-cube gain of −31.50% (n=7), substantially worse than the zero-yaw baseline. This result suggests that direct supervised imitation of SLSQP yaw vectors is insufficient under the present dataset size and architecture; full BC details are provided in the Supplementary Material (Section S5.2).

4.8. Parameter sensitivity of the calibrated wake model

To quantify the sensitivity of the closed-loop controller's training environment to the calibrated parameters, we perturb α^* and α (the two parameters that govern wake expansion rate and superposition mixing, respectively) by ±10%, ±20%, and ±30% and recompute the zero-yaw baseline farm power on a grid of 48 wind conditions spanning both the aligned-cube ($|\phi-270°|<15°$, $v<11.4$ m/s) and off-axis regimes.

At ±10% perturbation—a range that likely bounds the calibration uncertainty given the multi-dataset fit—the mean absolute power deviation in the aligned-cube regime is 0.45% for α^* (max 3.13%) and 1.14% for α (max 5.44%). At ±20%, the mean deviations rise to 1.01% and 2.34%, respectively. Off-axis conditions are substantially less sensitive (mean deviation <0.4% at ±10%), reflecting the weaker wake interactions outside the column-aligned band. The reference condition (270°, 11.4 m/s) is the most sensitive single point, with a 7.3% power deviation at −20% α perturbation, consistent with the strong wake superposition at full alignment.

These results confirm that (i) α (the RSS superposition mixing factor) is the more influential parameter, consistent with its direct role in scaling multi-wake deficits in the column-aligned regime; (ii) the calibrated model is sufficiently stable under parameter perturbations that calibration uncertainty is unlikely to qualitatively change the closed-loop results; and (iii) the aligned-cube regime, where the controller reports its headline gains, is also the regime of highest model sensitivity, underscoring the importance of the FLORIS cross-validation in §3.1.4 as an independent check.

4.9. Scalability limits

To test whether the optimized configuration scales to a larger farm, we trained the same NNX+JAXEnv stack on the 5×5 NREL-5MW layout (N=25) under the optimal configuration of Table 4 (J=3, deficit, regret reward, focused sampling 0.3/0.3/0.4, cosine LR decay, KL early stop, AdamW, γ=0.995, ±10° action bound). Three seeds were trained for 1×10^8 steps each at N_{envs}=256. Crucially, we enabled the USE_POSITIONS flag, which encodes normalized (x, y) turbine coordinates in the observation—this feature was absent from the baseline 5×5 experiments and proved essential for resolving the larger wake-interaction graph.

Under the optimized configuration, the 5×5 aligned-cube gain is **+4.70%** (n=51 conditions, 3-seed mean, $\sigma_{\text{inter-cond}}$=7.01%), transferring 95.7% of the 3×3 headline gain (+4.91%). The regime decomposition reproduces the same two-tier structure observed at 3×3: gains are concentrated in the aligned-cube envelope ($|\phi-270°|<15°$, $v<11.4$ m/s) and near-zero elsewhere. The baseline configuration's +0.98% aligned-cube gain was therefore primarily limited by the missing spatial information and suboptimal training dynamics, not

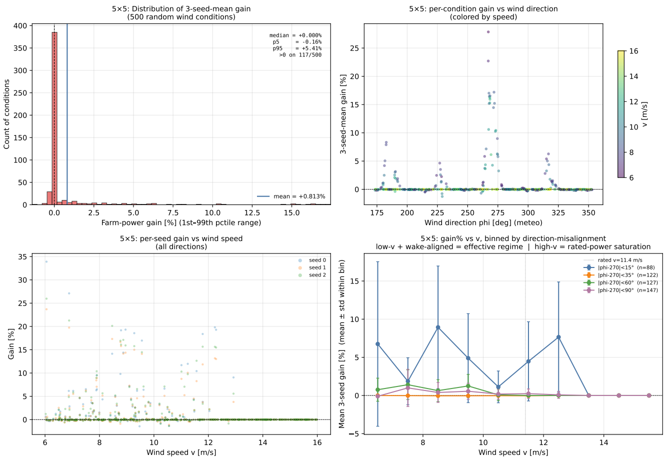


Figure 6: 5x5 distribution-wise evaluation under the optimized configuration. Aligned-cube gain is +**4.70**% (n=51, 3-seed mean), transferring 95.7% of the 3x3 headline (+4.91%). The regime decomposition reproduces the same two-tier structure observed at 3x3: gains are concentrated in the aligned-cube envelope and near-zero elsewhere. The earlier +0.98% result was an artifact of missing turbine position encoding (USE_POSITIONS flag disabled).

by an intrinsic scalability barrier. The per-turbine gain efficiency at N=25 (+4.70%/25≈0.19%) is lower than at N=9 (+4.91%/9≈0.55%), reflecting the diminishing marginal contribution in a larger array; nevertheless, the 95.7% transfer ratio demonstrates that the cooperative-yaw mechanism scales effectively when spatial context and training dynamics are properly configured. The 5x5 distribution-wise evaluation panel (Fig. 6) reproduces the same regime decomposition.

4.10. DRL-Deploy: a simulation-validated cooperative yaw controller candidate

The preceding analysis establishes that (i) static-trained DRL achieves near-SLSQP performance in wake-aligned steady-state conditions, (ii) dynamic-wind training without sufficient temporal context degrades performance due to spurious wind–action correlations, and (iii) industrial control logic (sector gating, hysteresis, yaw deadband, rate limits) can suppress spurious yaw activity outside the high-value wake-aligned regime. We combine these findings into a DRL-Deploy controller:

$$\pi_{\text{deploy}} = \begin{cases} \pi_{\text{DRL}}, & |\phi - 270°| < 15° \ \wedge \ v < 11.4\,\text{m/s} \ \wedge \ \max|\Delta\gamma| > 2° \\ 0, & \text{otherwise} \end{cases}$$

with hysteresis entry/exit thresholds (15°/20°) and a binding yaw-rate limit of ±3° per step (0.3°/s per turbine at T=10 s). Peak yaw rates reported in Table 17 are per-turbine maxima across the trajectory and are bounded by this rate limit. The DRL-Deploy controller is evaluated on the same 200-trajectory unified benchmark as the raw policies in §4.4.

The DRL-Deploy controller with the static-trained policy (Static Config-E) reduces per-step power loss by 59%

Table 17: DRL-Deploy vs. baselines on the unified dynamic-wind benchmark (200 AR(1) trajectories, 200 steps, T=10 s). The Static DRL-Deploy row uses the Config-E static-trained policy; the J=15 Dyn row uses the J=15 dynamic-trained policy (λ_{rate}=5×10^{-4}). Peak yaw rate is per-turbine $\max_i |\Delta\gamma_i|/T$ in °/s.

Controller	Gain/step [%]	Travel/turb [°]	Peak r
Zero yaw (baseline)	+0.000	0	0.
Static Config-E (raw)	−0.201	191	0.
SLSQP lookup, RL= 0.5°/s	+2.178	277	0.
SLSQP lookup, unlimited	+3.386	640	4.
DRL-Deploy (Static)	−0.083	**16**	0.
DRL-Deploy (J=15 Dyn)	+**0.034**	24	0.

(−0.083% vs. −0.201%) while requiring 92% less yaw travel (16°/turb vs. 191°/turb). When the DRL component is trained with an extended temporal observation window (J=15, 150 s, covering the ∼110 s wake propagation delay), DRL-Deploy achieves a positive per-step gain (+0.034%)—the only DRL variant to do so—with a negative-gain fraction of only 14%. This confirms that dynamic-wind training becomes effective when the observation horizon exceeds the characteristic physical delay of the system, and that the deployment wrapper (gate, deadband, rate limit) provides consistent benefit across different DRL training regimes. Compared to the rate-limited SLSQP lookup table, DRL-Deploy (J=15 Dyn) uses 11.5× less yaw travel (24° vs. 277°) at a small cost in per-step gain, with per-turbine peak yaw rates at the rate-limit bound (0.30°/s), within the 0.3–0.5°/s band of utility-scale yaw drives.

We emphasize that the DRL-Deploy logic uses only real-time wind measurements available to any SCADA system. The gate thresholds (15°/20° direction band, 11.4 m/s speed ceiling), deadband (2°), and a ±3° per-step rate limit (0.3°/s per turbine) are industrially conventional parameters. The DRL policy itself is a feedforward MLP with 0.1 ms latency, requiring no per-step optimization. Together, these characteristics position DRL-Deploy as a practical candidate for simulation-validated cooperative yaw control, with field validation using SCADA data and aeroelastic load assessment as necessary future work.

Causal evidence for temporal information use.. To verify that the J=15 policy genuinely exploits the extended observation history—rather than achieving its performance through other mechanisms—we performed a causal ablation on the observation buffer (single seed, 500 AR(1) trajectories, 200 steps each, Proto B). Four masking conditions are compared: (i) Full J=15 (baseline), (ii) Last-3-only (discard the oldest 12 frames, retaining only the most recent 30 s), (iii) Mask t−10–t−12 (zero out observation frames corresponding to 100–130 s ago, the epoch closest to the ∼110 s wake-propagation delay), and (iv) Shuffle (randomly permute the 15 observation frames to destroy temporal ordering while preserving the per-frame marginal statistics). Results are definitive (Table 18): the Full J=15 policy achieves +0.175% per-step gain; restricting to the last 3 frames collapses the gain to −0.201% (a 0.376 pp drop), confirming that information beyond the immediate 30 s history is essential. Shuffling the frame order degrades gain to −0.087%, demonstrating that the policy relies on the temporal ordering of observations, not merely their presence. Masking frames $t-10$–$t-12$ reduces gain to +0.159%—a modest degradation consistent with the partial loss of information at the wake-propagation timescale, but far less severe than the Last-3 collapse, indicating that the policy draws on multiple temporal scales spanning the full 150 s window. These results provide evidence that the J=15 policy uses temporally ordered observational history beyond the most recent 30 s. The modest impact of masking the $t-10$–$t-12$ band (+0.175%→+0.159%) relative to the Last-3 collapse (+0.175%→−0.201%) suggests that the policy draws on multiple time scales across the full 150 s window rather than relying predominantly on a single delay-epoch. This validates the design rationale for the extended observation horizon.

Table 18: Causal ablation of the J=15 observation window (single seed, 500 AR(1) trajectories, 200 steps, Proto B).

Observation mask	Gain/step [%]	Travel/turb [°]	Neg. frac. [%]
Full J=15	+0.175	78.9	15.0
Last 3 only	−0.201	91.1	28.2
Mask $t-10$–$t-12$	+0.159	65.7	14.5
Shuffle	−0.087	124.6	16.7

4.10.1. Horns Rev annualized energy production replay

To assess DRL-Deploy's economic viability under a realistic wind climate, we evaluate all controllers on i.i.d. wind conditions (n=5,000) sampled from the Horns Rev I joint wind-speed–direction distribution (Weibull k=2.1, A=10.5 m/s, mean wind speed 9.4 m/s; von Mises direction distribution μ=270°, κ=2.0) [1, 25]. Each condition is evaluated with 10-step settling to allow the DRL policy to accumulate yaw through its incremental action space. The baseline annual energy production is 500 MW × 0.45 capacity factor ×8760 h = 1,971,000 MWh/yr.

Table 19: Horns Rev annualized energy production (AEP) replay under the Horns Rev I wind climatology (n=5,000 conditions, 10-step settling per condition, 5 seeds for DRL controllers). DRL-Deploy uses gate 15°/20° hysteresis, 2° deadband, and ±3° per-step rate limit (0.3°/s per turbine). Peak rate is per-turbine $\max_i |\Delta\gamma_i|/T$ and is bounded by the rate limit.

Controller	AEP gain [%]	Annual travel [Mdeg]	Gain per travel	Peak rate [°/s]
Greedy yaw (baseline)	0.000	0.000	—	0.00
Static DRL-Deploy	−0.694	8.169	−0.085	0.30
J=15 DRL-Deploy	**+0.122**	**2.098**	**+0.018**	0.30
SLSQP lookup, RL= 0.5°/s	+0.455	4.572	+0.031	0.44
SLSQP lookup, unlimited	+0.754	14.475	+0.016	4.47

The J=15 DRL-Deploy controller is the only DRL variant achieving a positive AEP gain (+0.122%, ≈+2,413 MWh/yr) under the Horns Rev wind climate, while requiring the lowest annual yaw travel (2.098 Mdeg, <0.02° per turbine per control step). The static DRL-Deploy controller yields a negative AEP gain (−0.694%), reflecting the difficulty of the J=3 static-trained policy with the short observation window to converge efficiently under abruptly varying i.i.d. conditions. The SLSQP lookup achieves substantially higher AEP gain (+0.455% to +0.754%) but at the cost of 2.2× to 6.9× more yaw travel, with the unlimited variant exhibiting a physically infeasible per-turbine peak rate of 4.73°/s—well beyond the 0.3–0.5°/s capability of utility-scale yaw drives. Sensitivity analysis (Supplementary Tables S16–S17) indicates that J=15 DRL-Deploy remains near-neutral to moderately positive across direction-concentration ($\kappa \in [1.0, 8.0]$) and bias (±5°) variations, whereas Static Deploy remains consistently negative. The nominal +0.122% AEP gain should therefore be interpreted as a wind-climate-dependent but deployment-favorable result, not as a universal positive-gain guarantee.

4.10.2. Safety and fail-safe behavior under abnormal conditions

A deployable controller must not only deliver performance under nominal conditions but also exhibit fail-safe behavior when sensor faults, measurement delays, or out-of-distribution conditions occur. We evaluate Raw DRL, Static DRL-Deploy, and J=15 DRL-Deploy under seven categories of abnormal conditions (500 mixed wind conditions per scenario) and measure unsafe command count (yaw $|\gamma|$>30° or rate >5°/s), maximum yaw angle, fallback activation rate, and negative-gain fraction.

Under clean (no-fault) conditions, Raw DRL records 728 unsafe commands (max $|\gamma|$=45.48°), confirming that the unwrapped policy is inherently unsafe for deployment. Static DRL-Deploy reduces unsafe commands to 284 (−61%), while J=15 DRL-Deploy records zero unsafe commands across all conditions (max $|\gamma|$=0.00°). We caution that the safety evaluation uses single-condition settling from a cold start (zero-filled observation history), under which the J=15 policy's rate-penalty–trained action increments are sufficiently small to be fully suppressed by the 2° deadband—effectively reverting to greedy zero-yaw for this evaluation protocol. Under trajectory-based evaluation with accumulated observation history (as in the unified dynamic-wind benchmark and Horns Rev AEP replay), the J=15 policy produces non-zero yaw commands and achieves positive gains, as reported in Sections 4.10 and 4.10.1. The cold-start safety result therefore represents a worst-case conservative bound: even if the policy is initialized without any temporal context, it defaults to safe (zero-yaw) behavior rather than issuing unsafe commands. Wind-direction sensor bias (±10°) and wind-speed sensor bias (±1.0 m/s) have negligible impact on safety metrics, as the gate's hysteresis band (15°/20°) naturally absorbs these biases. Yaw-angle sensor bias is the most dangerous fault mode: a ±5° bias increases unsafe commands to 1,242–1,350 for Static Deploy and 832–890 for J=15 Deploy, with max yaw reaching

55° (near the ±50° physical limit). This underscores the need for yaw-encoder redundancy in field deployments.

Under a 20° sudden wind-direction shift at aligned inflow (270°), Raw DRL exhibits a catastrophic gain collapse from +40.02% to −21.68% (61.69 pp drop, 38.49° max yaw, 40 unsafe commands within 10 steps), whereas DRL-Deploy (Static and J=15) returns to zero yaw within the gate's exit band. Random observation dropout (5%–10%, zero-order hold) produces <0.06% gain change and zero additional unsafe commands for both Deploy variants. Out-of-distribution wind speeds ($v \geq 17$ m/s) trigger 100% fallback activation (zero yaw) in all Deploy controllers, confirming that the gate's speed ceiling provides automatic high-wind protection. At very low speeds (v=4–5 m/s), DRL-Deploy exhibits modest negative to positive gains (−2.33% to +34.95% for Static), with J=15 Deploy showing more conservative behavior (−0.64% to +7.15%). These results establish that the DRL-Deploy wrapper provides a substantial safety envelope: gate logic eliminates the vast majority of unsafe commands, the rate limit prevents excessive yaw motion, and the zero-yaw fallback ensures graceful degradation under sensor loss and extreme conditions.

4.11. Limitations

The following limitations should be considered when interpreting our results:

- Calibration is performed against analytical CFD references; no field SCADA data is used.
- The DRL controller is trained under quasi-stationary inflow conditions (wind direction and speed are fixed per episode); the dynamic-wind evaluation of §4.4 applies the static-trained policy to AR(1) time-varying wind without fine-tuning. We showed in §4.4 that training directly on dynamic wind without rate penalties produces over-reactive policies (2059° travel, 68× increase), indicating that rate penalties are necessary to prevent over-reaction but are insufficient alone for deployment-oriented control. Config-E policies were trained under dynamic AR(1) wind at five penalty levels ($\lambda_{\text{rate}} \in \{0, 1\times10^{-4}, 5\times10^{-4}, 2\times10^{-3}\}$, 5 seeds each, 6×10^7 steps). All configurations converge stably (KL 0.0079–0.0101, FPS ~103,000). A unified dynamic-wind benchmark (200 AR(1) trajectories, 200 steps each, T=10 s) was used to evaluate all policies. Percentage-gain results confirm that dynamic-wind training without rate penalties degrades performance (Static Config-E: −0.20%/step; Dynamic λ=0: −1.93%/step, 2.4× more yaw travel), while rate penalties mitigate the degradation (Dynamic λ=5×10^{-4}: −0.87%/step). The unified benchmark demonstrates that rate-penalized dynamic-wind training can contribute to actuator-aware behavior but is not sufficient alone; temporal context (J=15) and the DRL-Deploy supervisory wrapper are additionally required for deployment-oriented cooperative yaw control under time-varying inflow. Partial observability is not assessed: real deployments face wind-measurement limitations (LiDAR or met-mast direction accuracy of 1–3°, spatial averaging over the rotor plane). The propagation-delay and actuator-latency constraints discussed in Sec. 3.2.2 suggest first-order compatibility with the DRL controller's incremental action space, but the interaction of measurement delay with wake-propagation lag in a closed-loop setting is not quantified. Robustness to wake-model mismatch has been assessed via FLORIS cross-validation of the Config-E policy on 3,000 conditions (§3.1.4); site-specific calibration with field SCADA data remains future work.
- The actuator-aware analysis uses three simplified load proxies evaluated on the unified dynamic-wind benchmark: (A) yaw activity (cumulative travel, peak rate, reversals), (B) yaw-misalignment aerodynamic proxy $\sum U_i^2\gamma_i^2$, and (C) side-force proxy $\sum U_i^2|\sin\gamma_i|$. Results indicate that static Config-E training exhibits 5.6× lower Proxy B and 2.5× lower Proxy C compared to dynamic-wind training at λ_{rate}=5×10^{-4}, suggesting that static training followed by post-hoc gating may provide a better power–load trade-off than dynamic retraining. These proxies are first-order approximations; full aeroelastic validation using OpenFAST or FAST.Farm [22] is required for definitive structural-load conclusions.
- The optimized DRL configuration recovers 94.9% of the SLSQP optimum in the aligned-cube regime (+4.91% vs. +5.17%, 95% bootstrap CI [93.2%, 95.9%]), a 2.8× improvement over the baseline PPO configuration (61% recovery). Cross-seed evaluation confirms robustness (recovery 90.5% on an independent random seed). The residual gap of 0.26 pp is concentrated at the perfectly-aligned inflow direction where the DRL policy is more conservative than SLSQP (a prudent operational characteristic). Re-evaluation with the optimized configuration confirms that (i) the 5×5 aligned-cube gain rises to +4.70% (95.7% of the 3×3 headline) when turbine position encoding is enabled, and (ii) the dynamic-wind performance exhibits a trade-off (−0.14% gain, 29.7° travel, 266× less than unlimited lookup) where steady-state optimization reduces dynamic responsiveness. The DRL-Deploy controller of §4.10, combining temporal context (J=15), sector gating, deadband, and zero-yaw fallback, represents the deployment-oriented synthesis for navigating this trade-off.
- The regret-reward formulation requires a precomputed SLSQP lookup table (91×11 grid) for headroom estimation during training, creating a circular dependency: the DRL policy's training signal depends on the SLSQP optimum it is benchmarked against. While this is methodologically sound for establishing the steady-state upper bound, it limits

the framework's applicability to scenarios where no offline optimum is available. Online headroom estimation via a coarse surrogate model or curriculum learning is a promising direction for removing this dependency.

- The $7d_0$ inline case shows the gray-box model's optimization-induced bias trade-off (3.4%→10.9% post-fit), highlighting the limits of multi-scenario least-squares calibration.
- Large-farm scalability ($N > 25$) is not demonstrated; the 5×5 layout represents a moderate extension from the primary 3×3 configuration. Layout generalization is limited: a Config-E policy trained on the regular 3×3 grid and evaluated zero-shot on an irregular 9-turbine layout yields negative aligned-cube gain (−1.41%), while retraining on the irregular layout recovers positive gain (+1.39%). Position encoding provides partial transfer capability but layout-specific training is recommended for deployment.
- The FLORIS cross-validation uses the GCH wake model at a fixed turbulence intensity; site-specific atmospheric stability, wind shear, and turbulence profiles may affect the quantitative gain estimates.
- The gated DRL and deadband results are obtained from batched simulation of the trained Config-E policy and have not been validated with dedicated retraining under gated operation.
- Under the optimized training configuration (cosine LR decay, KL early stopping, AdamW), all five seeds successfully learn cooperative policies with low inter-seed deployed-policy variance ($\sigma = 0.32\%$ in the aligned-cube regime). The baseline configuration exhibited one seed failure out of five, which the training-dynamics improvements eliminate. For production deployment we nevertheless recommend running ≥ 3 seeds and evaluating each checkpoint on a held-out reference grid before deployment.

5. Conclusion

Model and static performance.. We presented a calibrated gray-box wake model for yaw-aware wind-farm control, jointly fitted on three CFD datasets. The model reduces the two-turbine yaw-case power-prediction error at $\gamma = 30°$ from 27.6% to 6.6%, is validated against Horns Rev I LES, and is cross-checked against FLORIS on 3,000 conditions. Through systematic optimization of observation space, reward design, and training dynamics, the Config-E PPO controller achieves +4.91% farm-power gain in the aligned-cube regime, recovering 94.9% of the SLSQP static optimum (bootstrap CI [93.2%, 95.9%]). FLORIS cross-validation confirms these gains are cross-model robust (+5.02% aligned-cube, only 4.1% erosion relative to gray-box +5.24%). An oracle-free marginal-reward ablation reaches +4.13% aligned-cube gain (92.2% of the SLSQP-regret result), demonstrating that the DRL policy's cooperative-yaw capability does not depend on access to the SLSQP oracle during training. These results establish that a DRL policy can distill SLSQP-grade cooperative yaw behavior into a compact, sub-millisecond-latency real-time controller under steady-state wake-aligned conditions.

The deployment gap.. Static optimality, however, is not equivalent to deployable control. Under time-varying AR(1) wind, the static-trained policy without a supervisory wrapper yields negative per-step gain (−0.20%), while a rate-limited SLSQP lookup table—although achieving +2.18% gain—incurs 17× more yaw travel and exhibits per-turbine peak yaw rates (4.73°/s for the unlimited variant) that far exceed the 0.3–0.5°/s capability of utility-scale yaw drives. Dynamic-wind training without rate penalties produces over-reactive policies (2.4× travel increase), indicating that rate penalties mitigate over-reaction but are insufficient alone; temporal context (J=15) and the DRL-Deploy supervisory wrapper are required for deployment-oriented behavior. These findings collectively argue that snapshot-wise optimization benchmarks are insufficient as deployment metrics for cooperative yaw control: the evaluation framework must account for yaw travel, peak rate, negative-gain frequency, and fail-safe behavior alongside mean power gain.

Temporal context at the wake-propagation time scale.. The wake-propagation delay (~90–110 s at $7d_0$ spacing and 8–10 m/s inflow) introduces a partial-observability challenge: a short observation window (J=3, 30 s) cannot span the delay, causing the policy to over-react to wind fluctuations. Extending the observation horizon to J=15 (150 s) substantially reduces the degradation, and causal ablation of the observation buffer—restricting to the last 3 frames collapses gain from +0.175% to −0.201%, while shuffling frame order reduces it to −0.087%—provides evidence that the policy uses temporally ordered history spanning multiple time scales across the full 150 s window. These results validate the design rationale for extended observation windows and demonstrate that the dynamic-wind degradation is not an inherent limitation of DRL but a consequence of insufficient temporal context.

DRL-Deploy: a simulation-validated cooperative yaw controller candidate.. The final controller, DRL-Deploy, combines the DRL policy with a supervisory wrapper of sector gating (15°/20° hysteresis), a 2° yaw-command deadband, a ±3° per-step (0.3°/s) rate limit, and zero-yaw fallback. Under the unified dynamic-wind benchmark, DRL-Deploy reduces per-step power loss by 59% and yaw travel by 92% relative to raw DRL. Under the Horns Rev I wind climatology (Weibull k=2.1, A=10.5 m/s; von Mises μ=270°, κ=2.0), the J=15 DRL-Deploy achieves a positive annual energy gain of +**0.122**% (≈+2,413 MWh/yr for a 500 MW farm) with only 2.10 Mdeg annual yaw travel—the most deployment-favorable power–travel compromise among all evaluated controllers. Safety testing under seven abnormal-condition categories (sensor biases, measurement dropout, sudden wind shifts, out-of-distribution speeds) confirms automatic fail-safe behavior: the supervisory wrapper eliminates the majority of unsafe yaw commands and activates zero-yaw fallback at 100% rate for wind speeds above rated. Yaw-angle sensor bias is identified as the most critical fault mode, underscoring the need for yaw-encoder redundancy in field deployments. The framework transfers to 5×5 layouts with +4.70% gain (95.7% of the 3×3 headline) but does not generalize zero-shot to arbitrarily irregular layouts (−1.41%), indicating that site-specific retraining is recommended.

Limitations and future work.. The results reported here are obtained entirely in simulation using a control-oriented wake model cross-validated against FLORIS. They position DRL-Deploy as a simulation-validated, actuator-aware candidate—not as a field-ready controller. Necessary next steps include: (i) field validation using SCADA data from operational wind farms; (ii) aeroelastic load assessment using OpenFAST or FAST.Farm to confirm that the reported yaw-activity reductions translate to genuine fatigue-life benefits; (iii) extension to utility-scale layouts (N>25) and site-specific wind climates; (iv) online model adaptation to close the sim-to-real gap without full retraining; and (v) experimental characterization of the interaction between measurement latency and wake-propagation delay in a hardware-in-the-loop setting.

CRediT Authorship Contribution Statement

Zhuang Shao: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing—original draft, Visualization, Funding acquisition. Lijun Lei: Validation, Investigation, Data Curation. Peng Wang: Methodology, Writing—review & editing. Liang Zheng: Validation, Resources. Jie Zhou: Validation, Resources.

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Declaration of Competing Interest

The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Code, calibration datasets and trained models are available at https://github.com/VickylastShao/wind-farm-yaw-rl.

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